

RESEARCH ARTICLE

Structural and functional development of twelve newly established floodplain pond mesocosms

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Abstract

Ecosystems are complex structures with interacting abiotic and biotic processes evolving with ongoing succession. However, limited knowledge exists on the very initial phase of ecosystem development and colonization. Here, we report results of a comprehensive ecosystem development monitoring for twelve floodplain pond mesocosms (FPM; 23.5 m × 7.5 m × 1.5 m each) located in south-western Germany. In total, 20 abiotic and biotic parameters, including structural and functional variables, were monitored for 21 months after establishment of the FPMs. The results showed evolving ecosystem development and primary succession in all FPMs, with fluctuating abiotic conditions over time. Principal component analyses and redundancy analyses revealed season and succession time (i.e., time since ecosystem establishment) to be significant drivers of changes in environmental conditions. Initial colonization of both aquatic (i.e., water bodies) and terrestrial (i.e., riparian land areas) parts of the pond ecosystems occurred within the first month, with subsequent season-specific increases in richness and abundance for aquatic and terrestrial taxa over the entire study period. Abiotic environmental conditions and aquatic and terrestrial communities showed increasing interpond variations over time, that is, increasing heterogeneity among the FPMs due to natural environmental divergence. However, both functional variables assessed (i.e., aquatic and terrestrial litter decomposition) showed opposite patterns as litter decomposition rates slightly decreased over time and interpond differences converged with successional ecosystem developments. Overall, our results provide rare insights into the abiotic and biotic conditions and processes during the initial stages of freshwater ecosystem formation, as well as into structural and functional developments of the aquatic and terrestrial environment of newly established pond ecosystems.

KEYWORDS

ecosystem development, ecosystem function, freshwater colonization, primary succession, principal component analysis

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TAXONOMY CLASSIFICATION

Agroecology; Biodiversity ecology; Ecosystem ecology; Population ecology

1 | INTRODUCTION

Ecosystems are characterized by abiotic and biotic components that are linked by multiple processes and through nutrient cycles and energy flows. These systems are dynamic and developing entities controlled by external and internal factors. Mature (climax) ecosystems, developed over long-term periods, are often complex and relatively stable systems. In contrast, ecosystems at their initial developmental stages are far less complex and stable, with fewer biotic and abiotic components interacting, and characterized by high levels of stochasticity (Buma et al., 2019; Fath et al., 2004). The period between the start of ecosystem development (i.e., the “point zero”) and the quantitative establishment of a first dynamic equilibrium of element cycling can be defined as initial ecosystem stage (Elmer et al., 2013; Schaaf et al., 2011). Such initial systems are more homogenous and less structured than mature ecosystems, which are the results of previous ecosystem succession (Elmer et al., 2013; Odum, 1969). Generally, succession as an important concept in ecology encompasses the directional and continuous occurrence of a range of successional sequences within an ecosystem over varying time scales (Begon et al., 2006). With ongoing succession, the ecosystem complexity increases, with “opportunistic” pioneer, short-lived species colonizing the ecosystem during initial stages (Connell & Slatyer, 1977), which are replaced at later stages by new erratic species with more passive colonization abilities (Ruhí et al., 2013). In addition to changes in community composition, also biomass, interactions, respiration, and information (e.g., genetic diversity) increase during ecosystem development, whereas entropy decreases (Fath et al., 2004; Odum, 1969).

Natural ponds as abundant and ecologically important freshwater ecosystems in Europe and globally (Downing et al., 2006; Kristensen & Globevnik, 2014) have been in the focus of succession research for more than a century (e.g., Allee, 1911). One prominent example is the Pinkhill Meadow pond complex, consisting of approximately 40 permanent, semipermanent and seasonal ponds in the River Thames Floodplain, Oxfordshire, UK. Selected ponds of this complex have been monitored since their construction in 1990 (Freshwater Habitats Trust, 2021). However, the knowledge on the abiotic conditions, biotic communities, and initial ecological processes, including their drivers, of newly established pond sites is generally limited (Miguel-Chinchilla et al., 2014; Williams et al., 2008). It is rarely possible to study the “point zero” of ecosystem development and primary succession in newly established ecosystems under natural conditions (Schaaf et al., 2011). Most ecological research is conducted in established, climax pond ecosystems, without investigating the abiotic and biotic conditions and processes that led from the initial stages to the development of mature systems (Raab et al., 2012). Also, available information on key environmental

factors, such as physico-chemical water quality parameters and seasonality, are limited leaving a knowledge gap in what is driving primary succession and ecosystem development in newly established ponds (Williams et al., 2008). For this reason, monitoring of pond ecosystem development from the “point zero” provides valuable information on factors driving ecosystem structuring and functioning at initial development stages.

The colonization and subsequent community development during primary successional stages in new pond habitats is determined by stochastic primary effects, local environmental conditions, and biological interactions (Buma et al., 2019; Weidlich et al., 2021; Wiegler et al., 2017). Along the initial structural developments, also ecosystem functioning evolves driven by the functional traits of the organisms present. The differences in aquatic taxa composition directly affect ecosystem processes and consequently functioning (de Bello et al., 2010; Herbert et al., 2016). Ecosystem functions such as litter breakdown (Gessner & Chauvet, 2002) also denote suitable indicators for the primary successional development and integrity of young emerging ecosystems.

In this study, we conducted an ecosystem primary succession monitoring considering a variety of physico-chemical parameters, aquatic and terrestrial taxa as well as variables to assess ecosystem functioning. This was done to investigate pond ecosystem developmental trajectories during 21 months from their establishment. The monitoring was conducted in a system of twelve natural floodplain pond mesocosms (FPM 1–12, natural and undisturbed lotic freshwater ecosystems planned and constructed at a research site and operated without anthropogenic manipulation) that have been established in 2017 at the EuBerthal Ecosystem Research Station (EERES; see <https://www.uni-koblenz-landau.de/de/landau/fb7/umweltwissenschaften/eeres>) in south-western Germany. The FPMs used here represent systems at “point zero” of ecosystem development and enable repeated sampling and analyses of initial development stages in a fully replicated way. We used ordination methods (i.e., principal component analyses [PCA] and redundancy analyses [RDA]) to analyze pond environmental conditions and inter-pond variations during initial successional stages.

Besides insights into primary succession and initial pond ecosystem developments, our study contributes to the knowledge of experimental mesocosm studies used to disentangle stressor effects from other confounding factors (e.g., Finnegan et al., 2018; Hua & Relyea, 2014), as aquatic communities of these systems also generally are in very early stages of succession. Insights into primary succession of pond mesocosms, factors driving aquatic community development, and lotic ecosystem development trajectories as gained here may benefit the interpretation of findings from other studies focusing on artificially stressed mesocosm experiments. The objectives of this paper are (1) to monitor and describe the ecosystem development of

twelve FPMs using 20 physico-chemical and biotic parameters; (2) to disentangle and analyze factors driving the development of environmental conditions within the pond ecosystems; and (3) to ascertain interpond variations (i.e., differences among ponds by natural heterogenization) over time.

2 | METHODS

2.1 | Study site and FPMs

This study has been conducted at the EERES located in Rhineland-Palatinate, south-west Germany (49°15'14"N, 7°57'42"E). The EERES site is situated in a small valley of the Franco-German Palatinate Forest-North Vosges UNESCO biosphere reserve, which is an extensively forested low-mountain range. In 2017, twelve FPMs have been constructed at the EERES site (see Figure S1 for an overview). Each FPM is connected to a small stream called Sulzbach via controllable inlets and overflows. The inlets, except for rare events (see below), were kept closed during the study period preventing the water from the Sulzbach to mix with pond water. The Sulzbach belongs to the stream type "small, fine-substrate dominated, siliceous highland rivers" (Westermann et al., 2011). The upstream catchment of the Sulzbach has only minor anthropogenic influences and the stream is characterized by a high structural and ecological quality. The FPMs are orientated in east (inflow)–west (outflow) direction. Each FPM has a dimension of 23.5 m × 7.5 m with V-shaped banks at three sides and a flat water–land floodplain area at the inflow (Figure S2a). The depths of the FPMs increase towards the outflow area (bed slope 1:20), providing a maximum water storage level of 1.5 m. The banks and the bottoms are impervious to water due to a PVC membrane covered by a sand layer of 15 cm; an additional 0.5 m gravel layer (uniform gravel diameter of >2 cm) covers half of each FPM pond bottom toward the outflow area (Figure S2a). The upper half of the banks was seeded with a standard grass mixture (*Festuca rubra* ssp., *Festuca trachyphylla*, *Poa pratensis*) in October 2017 to prevent erosion. After flushing the newly constructed FPMs twice with water from the Sulzbach, the ponds were filled with Sulzbach water on 04 November 2017 to a water level of 30 cm at the outlet sites (Figure S2b). This water level was kept constant at 30 cm (±10%) during the entire study period, meaning that losses due to evaporation were compensated by refilling with water from the Sulzbach. However, refilling of the FPMs was needed rather infrequently (i.e., approximately every 2–3 months depending on meteorological conditions). During the refilling, in order to ensure reproducibility, the inlets of all FPMs were always opened at the same time and for the same brief (i.e., approximately 2 h) duration. Given that the volume of the inflow water was small compared with the water volume of the ponds and that all FPMs received water from the same stream, we estimate the influence of the stream water on the abiotic environmental conditions of the FPMs as marginal. Apart from refilling the ponds, no water inflow into the FPMs occurred. The FPMs thus represent small lentic water bodies with water–land floodplain areas.

It has to be noted that the water from the Sulzbach was not sterilized before it was used to initially fill or to refill the ponds. However, during the entire experimental phase, no anthropogenic manipulation or management of the twelve FPMs took place to allow natural primary succession and undisturbed development. The EERES research station is completely fenced off to avoid disturbances by human visitors or larger wild animals. Since October 2019, the FPMs are a key research site of the DFG Research Training Group 2360 "SystemLink" (<https://systemlink.uni-landau.de>).

2.2 | Overview of the monitoring program and sampling methods

A monitoring program was launched immediately after the establishment of the FPMs. The initial ecosystem development and primary succession of the twelve FPMs from a defined "point zero" was monitored for 21 months from 06 November 2017 until 31 July 2019. Overall, 20 variables were surveyed: four parameters describing the general conditions at the EERES site and of the FPMs; five physico-chemical water quality parameters; nine biotic parameters; two functional parameters. Together, the 20 variables surveyed describe both the aquatic and adjacent terrestrial FPM ecosystem development (Table S1). Because of the high workload, it was not feasible to achieve the same temporal coverage and sampling dates for all parameters during the monitoring; not all parameters were surveyed regularly and during the entire study period. Specifically terrestrial species (i.e., terrestrial vegetation, ground beetles, leafhoppers) were assessed at a few distinct sampling dates only (see details on sampling design below). For this reason, absence of values in the figures and tables correspond to absence of data. During sampling campaigns, all sampling and measurement equipment was completely rinsed with tap water after usage in one FPM to prevent the transfer of organisms or biological material between the ponds. In addition, all sampling and measurement equipment (e.g., polyethylene mesh bags, data logger) were leached for at least 24 h in tap water (with at least three water exchanges) before it was placed in the FPMs in order to prevent leaching of chemicals into the water.

The information on meteorological conditions, soil temperature, and water level fluctuations can be found in the Appendix S1 (supporting results). To track visible real-time changes during the FPM development, we recorded a time-lapse video of FPM9 using a daily camera trap (Bushnell Trophy Cam HD) (available at <https://youtu.be/vfTNpwyYhFE>).

2.3 | Physico-chemical water quality parameters

Water samples were taken monthly or bimonthly in each FPM 5–10 cm below the water surface using high-density polyethylene bottles. Water samples were analyzed by an accredited laboratory applying standard analytical methods (AGROLAB, Germany) for the following components: total organic carbon (TOC), dissolved

organic carbon (DOC), fluoride, chloride, nitrite, nitrate, phosphate, sulfate, ammonium, sodium, potassium, calcium, and magnesium. The limits of quantification (LOQ) are given in Table S5. Please note that due to analytical problems (i.e., equipment failure), not all components could be analyzed at all sampling dates. Detected concentrations (i.e., concentration > detection limit) less than the LOQ were replaced by $0.5 \times \text{LOQ}$ for data evaluation and visualization.

Specific conductivity and pH were measured monthly or weekly 5–10 cm below the water surface near the outlet sites of the FPMs using a portable analytical device (WTW Multi 3630 IDS). No data for conductivity are available for January 2019.

Water temperature and dissolved oxygen were recorded by a miniDOT® logger (miniDOT® USB Oxygen Logger 7392) installed 5 cm above the bottom gravel layer (i.e., at 25 cm water depth) near the outlet site of each FPM. The miniDOT® logger recorded the water temperature and dissolved oxygen with a resolution of 10 min; daily averages of water temperature and dissolved oxygen were used for evaluations. Monitoring results for all physico-chemical water quality parameters are presented in the Appendix S1 (supporting results).

2.4 | Aquatic macroinvertebrates

Benthic macroinvertebrates were sampled 15 times over the sampling campaign using pebble baskets as artificial substrates (see e.g., Brock et al. (1992) for details). Pebble baskets consist of a polyethylene mesh (length 15 cm; width 15 cm; height 7.5 cm) formed as a pyramid and were filled with the bottom substrate (gravel with a diameter of >2 cm) of the FPM. While the upper part of the pebble basket was made of a coarse mesh (2 cm aperture) to enable easy colonization, the lower section was made of a fine mesh of 0.5 cm aperture to prevent escaping of macroinvertebrates during sampling. In each pond, a pebble basket was placed at the bottom near the inlet and outlet sites, as well as in the middle of each FPM, given a total of three baskets per FPM. Before animal collection, baskets were left for colonization for three weeks. At the end of each sampling campaign, the pebble baskets were gently retrieved from each FPM with a dip net (500 µm mesh aperture). The macroinvertebrates present on the pebble basket substrates were removed and preserved in ethanol (70% vol.). All individuals were subsequently identified to the family level.

2.5 | Merolimnic insect emergence

We sampled the emergence of merolimnic insects from April to July 2019 using three floating pyramidal tents with a basal area of 0.25 m² as emergence traps (Cadmus et al., 2016). The emergence traps were made of nylon mesh and anchored to the pond bottom. Emergence traps were installed from 24 April 2019 until 31 July 2019 and emptied once per week, that is 15 times in total. Three

emergence traps per pond were placed at the water surface at locations comparable to the pebble basket sites. Emerged insects were collected in plastic bottles connected to the top of the trap and filled with ethylene glycol to preserve insects. Upon collection, aquatic insects were stored in ethanol (70% vol.) and subsequently identified to the family level. All results of the merolimnic insect emergence monitoring are available in the Appendix S1 (supporting results).

2.6 | Zooplankton

Zooplankton was sampled monthly in each FPM with a plankton net (20 cm diameter; 65 µm mesh aperture). Overall, 15 samples were taken in all FPMs during the study period, with no samples available for January and February 2018. For sampling, the plankton net was moved gently across 10 m horizontally through each FPM at a distance of approximately 1 m from the banks. After retrieval, samples were rinsed through a net (mesh aperture 120 µm) and stored in ethanol (70% vol.). Zooplankton abundances (individuals >250 µm) were assessed by the Institute for Fishery of the Bavarian State Research Center for Agriculture using the ZooScan V4 system (HydroptiC, version 2.4.0), a digital zooplankton image analysis (Gorsky et al., 2010). Please note that zooplankton individuals <250 µm could not be assessed by the ZooScan digital image analysis.

2.7 | Amphibians (tadpoles) and submerged vegetation

Amphibians and submerged vegetation, that is, filamentous algae and *Elodea spec.*, were assessed by visual inspection of the percentage tadpole coverage and the percentage submerged vegetation coverage, respectively, of the water surface area. Visual inspection for tadpole presence was conducted in spring and early summer (i.e., April–June) 2018 (four times) and 2019 (16 times, observations from end of March until June), that is, during times with tadpole presence. For submerged vegetation, visual inspection was made 30 times at bimonthly or monthly intervals during the study period beginning on 31 January 2018. No distinction could be made regarding living and dead submerged vegetation. During each visual inspection, the presence of tadpoles and submerged vegetation was recorded manually using a layout template; the layout templates were subsequently digitized and the percentage areas covered by tadpoles and submerged vegetation, respectively, were subsequently analyzed by image recognition software (Adobe Photoshop version 21.2). In brief, we determined the total number of pixels for the entire surface area of each FPM and the pixel number of the areas covered by tadpoles or submerged vegetation. The percentage areas covered were then calculated by dividing the number of pixel of areas covered by tadpoles or submerged vegetation by the total number of pixels of the entire surface area of the respective FPM.

2.8 | Crayfish and fish

The presence of crayfish, which may dominate ecosystem development (e.g., Reynolds et al., 2013), in the FPMs were monitored using three baited standard crayfish traps (type: "Pirate") per FPM. Overnight monitoring was conducted monthly between April and October in 2018 and between March and July in 2019. The presence of fishes was also checked in the crayfish traps, as well as by visual inspection during each visit of the FPMs. No crayfish or fish were found during the entire study period.

2.9 | Terrestrial vegetation

The terrestrial vegetation was recorded twice by visually inspecting the northern and southern banks as well as the terrestrial part of the floodplain area of each FPM. The vegetation surveys took place in November 2018 and May 2019.

2.10 | Ground beetles

Ground beetles were collected by pitfall traps (Schirmel, 2020) during seven sampling campaigns conducted in late spring and autumn 2018, as well as in late spring 2019. One pitfall trap was placed at the northern and southern bank of each FPM and two pitfall traps at a distance of two meters from the shore into the terrestrial part of each floodplain area. The pitfall traps consisted of a plastic cup (6.5 cm diameter) filled with ethylene glycol diluted 1:3 with water and a drop of detergent. At the end of each sampling campaign, the pitfall traps' contents were sieved and the ground beetles were stored in ethanol (70% vol.). Ground beetle individuals were subsequently identified to species level.

2.11 | Leafhoppers

Leafhoppers were surveyed on 26 June and 21 September 2019 at the northern and southern banks of FPM3, FPM5, FPM8, FPM10. Leafhopper collection was conducted by suction sampling with a modified leaf blower (SH 56 Stihl, Waiblingen, Germany) using 80 suction pulses per sampling (Kormann et al., 2015). The leafhopper catches were transferred into a bucket and subsequently separated from the rest of the catch with an aspirator. Leafhopper samples were then stored in ethanol (70% vol.) and identified to species level.

2.12 | Aquatic and terrestrial litter decomposition

Aquatic leaf litter decomposition was approximated using the litter-bag method (Benfield et al., 2017). In brief, coarse- (10 mm aperture) and fine-mesh (1 mm aperture) litterbags were filled with approximately 4 and 2 g (weighed to the nearest 0.001 g), respectively, of oven-dried (at 60°C for 24 h) alder leaves (*Alnus glutinosa* (L.) Gaertn.)

(Voß et al., 2015). Fine-mesh and coarse-mesh bags were employed in the FPMs to separate the microbially and the shredder-mediated share of leaf litter decomposition. One coarse-mesh and one fine-mesh litterbag were submerged near the inlet and outlet sites, as well as in the middle part of each FPM for 20–25 days at nine dates between May 2018 and June 2019. After retrieval, the remaining leaf material was gently rinsed under running water to remove mineral particles and macroinvertebrates, oven-dried at 60°C for 24 h and reweighed to the nearest 0.001 g.

The litter decomposition rate k was calculated for each FPM and the respective sampling dates using the following formula (Benfield et al., 2017):

$$k = \frac{-\ln\left(\frac{S(t)}{S(0)}\right)}{t}$$

where $S(t)$ is the leaf mass as a function of deployment time t and $S(0)$ is the initial mass of the coarse- and fine-mesh litterbags, respectively. The mass data from the fine litterbags were used to calculate microbial decomposition ($k_{\text{microbial}}$). For the calculation of the litter decomposition by shredders ($k_{\text{shredders}}$), $S(t)$ of the coarse litterbag at a given sampling site was corrected for the mass losses due to microbial decomposition by adding to $S(t)$ the amount degraded over time in the coarse litterbags (i.e., $S(0) - S(t)$) multiplied by the mean percentage microbial decomposition of the respective fine litterbags. For data evaluation, decomposition rates were averaged across the three sampling locations of a given pond.

The terrestrial litter decomposition was assessed using the tea bag approach (Keuskamp et al., 2013). In brief, one green tea and one rooibos tea bag (brand "Lipton") were buried pairwise at a depth of five cm at the middle of each the north and south banks (50 cm above water surface) of each FPM for approximately 21 d at twelve dates during the study period. After retrieval, adhered soil particles were removed and the filling of the tea bags was oven-dried for 24 h at 60°C and weighed to the nearest 0.001 g. The terrestrial litter decomposition was calculated as linear weight loss per day (mg/d) for both green and rooibos tea as follows:

$$\text{Linear weight loss per day} = \frac{(M_R - M_t)}{t}$$

where M_R is the reference mass derived from averaging the weight of ten green and rooibos tea bag fillings, respectively, and M_t is the mass of the green and rooibos tea bag fillings after deployment time t . The mean value of the north and south bank sampling sites of a given FPM was calculated separately for green and rooibos tea bags, respectively, and used for data evaluation.

2.13 | Generalized additive models for temporal trend analyses

To visualize and assess overall temporal trends in biotic communities and ecosystem function, we fitted generalized additive models

(GAM) using the mgcv R package (Wood, 2011). We fit models for the following variables across all FPMs as functions of time since establishment: the number of macroinvertebrate families, the number of zooplankton individuals per sample, the percentage coverage by submerged vegetation, the aquatic litter decomposition rates $k_{\text{microbial}}$ and $k_{\text{shredders}}$, and the terrestrial litter decomposition expressed as linear weight loss per day for green tea and rooibos tea. In most cases, we used a Gaussian residual distribution and ten basis functions. For the two models on the number of macroinvertebrate families and the number of zooplankton individuals, we used a negative binomial residual distribution and for the latter 19 basis functions. For the model on $k_{\text{microbial}}$ and $k_{\text{shredders}}$, we used nine and three basis functions, respectively. Residual diagnostics were performed visually and the number of basis functions was evaluated with the test proposed in Wood (2017), available through the *gam.check()* function.

2.14 | PCA and RDA of pond ecosystem developments

We used PCA and RDA ordination analyses to analyze aquatic ecosystem development and environmental conditions of the twelve FPMs and to assess interpond environmental diversities. Briefly, PCA allows to reduce the dimensionality of the data set of environmental variables by explaining the correlation among a large number of environmental variables in terms of new orthogonal, uncorrelated variables (principal components [PCs]) without losing much information (Olsen et al., 2012; Ramette, 2007). Using multiple linear regression, RDA extends PCA by explaining variation between independent and dependent variables within an iterative process to find the best ordination (Ramette, 2007). The following environmental variables (month scale) were included in the PCA and the RDA: specific conductivity, pH, dissolved oxygen, water temperature, DOC, fluoride, chloride, nitrate, phosphate, sulfate, sodium, ammonium, potassium, magnesium, calcium, water level, and submerged vegetation as an indicator of the habitat structure of the FPMs (Caquet et al., 2001; Christman et al., 1994). All other environmental variables were excluded as they were either not directly related to the analysis of the aquatic ecosystem development and pond environmental conditions (e.g., terrestrial vegetation) or had an excess in number of missing values (e.g., tadpole coverage, leaf litter decomposition) across the entire study period (see Table S13 for a complete list of excluded variables). We conducted full PCAs and RDAs, respectively, for the entire study period, as well as separate PCAs and RDAs for each year, that is, for December 2017–June 2018 (first year), and for December 2018–June 2019 (second year); data for noncomparable months were excluded from PCAs for individual years. PCAs and RDAs were conducted both for months/seasons for temporal primary succession analyses, as well as using grouping of ponds to ascertain pond environmental diversities. We used stepwise forward selection in RDA and conducted additional variance partitioning analyses (VPA; e.g., Dray et al., 2012). Data treatment

included removing of n/a, checking and removing of outliers, as well as logarithmic ($\log_{10} + 1$) data transformation. Assumptions (e.g., linearity, homogeneity of variances, multicollinearity, residuals) for PCA and RDA were checked prior to analyses and selection of PCA axes was done using the broken stick approach (MacArthur, 1957). All statistical analyses were done using R (version 4.0.2).

3 | RESULTS

3.1 | Colonization of aquatic and terrestrial ecosystem compartments

3.1.1 | Aquatic macroinvertebrates

Overall, 7998 individuals belonging to 35 macroinvertebrate families and seven orders were identified in the twelve FPMs during the study period. The temporal trajectory for family richness shows increasing richness from December 2017 (mean of 0.7 families; 95% CI [0.24, 1.09]) until the end of August 2018 with a mean of 6.8 (95% CI [5.82, 7.68]) families across the FPMs (Figure 1a). After a decrease in family richness until October 2018, family richness leveled off until the end of the study period at around a mean of five families per FPM. GAM indicated a significant temporal trend for family richness development across all FPMs ($p < .001$; Figure 1a). However, there were large variations in family richness among FPMs at the individual sampling dates, particularly from summer 2018 onwards (Figure 1a). A maximum of 11 macroinvertebrate families per individual sampling was recorded for FPM12 (August 2018) and FPM10 (January 2019).

Variation was found also among ponds, across the study period; most of the macroinvertebrate families ($n = 19$) were found in FPM12 and 17 families in FPM1, FPM3, FPM6, FPM7, FPM9; in contrast, only eleven families were identified in FPM11 (Table S7). Concerning abundances, most individuals were found in FPM3 ($n = 1052$) and FPM12 ($n = 1494$), whereas FPM5 ($n = 332$) had the lowest abundances (Table S7).

3.1.2 | Zooplankton

The trajectory of zooplankton population size shows fluctuations over the study period. Zooplankton population size in the FPMs was low (i.e., <100 individuals per sample) until the end of June 2018 (Figure 1b; Table S7). Abundances subsequently increased reaching a peak at the end of August 2018 with an average of 1852 (95% CI [320, 3384]) individuals per sample across all FPMs. After a subsequent decrease to a minimum in January 2019 (mean of 100 (95% CI [43, 157]) individuals per sample across FPMs), zooplankton abundances increased again to a maximum in mid-2019 (mean of 2607 (95% CI [1585, 3628]) individuals per sample across FPMs). GAM analysis indicated the overall temporal trend of zooplankton population sizes in the FPMs to be significant ($p < .001$; Figure 1b).

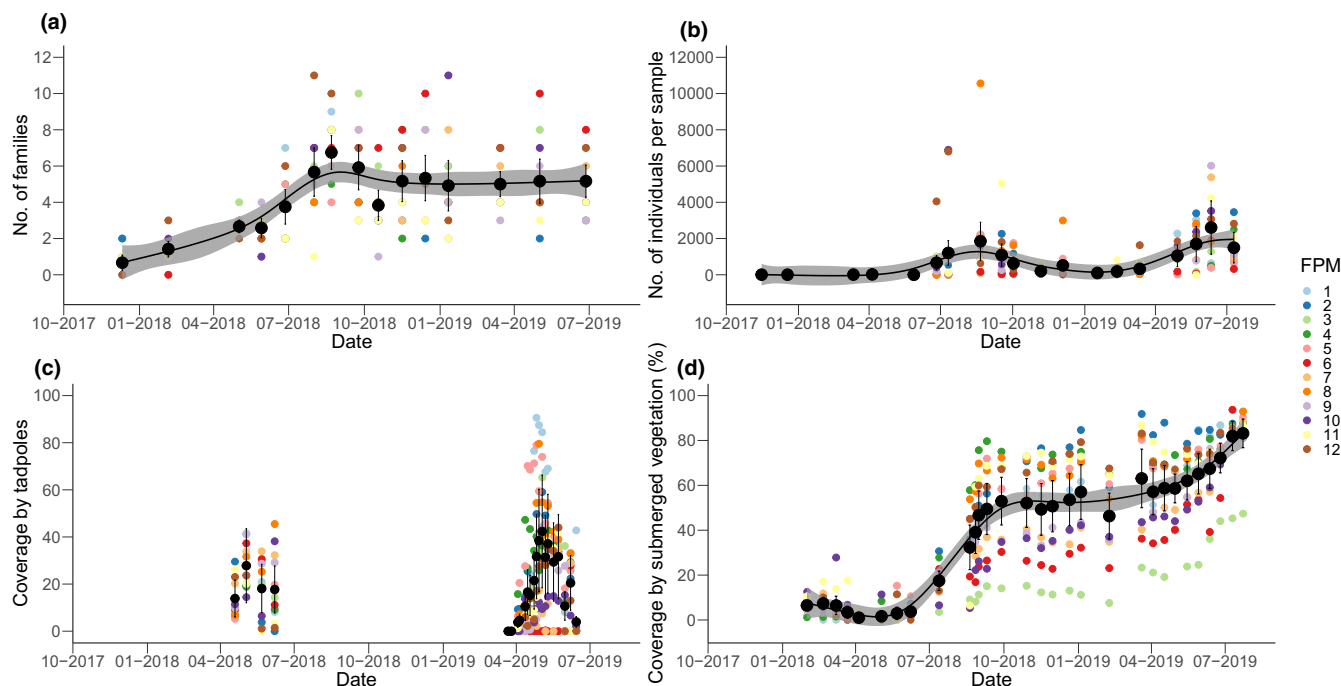


FIGURE 1 Generalized additive model (GAM) plots for (a) number of macroinvertebrate families and (b) zooplankton population (no. of individuals per sample) in the FPMs over time. (c) Percentage coverage of FPM 1–12 by tadpoles in spring/summer 2018 and 2019, and (d) GAM plot for percentage coverage of the FPMs by submerged vegetation over time. Black circles depict means and error bars show 95% confidence intervals. The gray shaded areas in (a), (b) and (d) indicate the 95% confidence intervals

Considering individual ponds, large differences in zooplankton abundances were found between FPMs at individual sampling dates specifically during late spring and summer months, while differences were clearly smaller in autumn and winter months (Figure 1b). In August 2018, FPM8 showed an exceptionally high abundance of 10,560 individuals per sample. Zooplankton individual numbers were overall highest in FPM2 and FPM8 to FPM12, whereas FPM3 and FPM6 generally had lowest abundances (Table S7).

3.1.3 | Amphibians (tadpoles)

Irrespective of the year of observation, tadpoles were present only from April until June. In 2018, tadpole coverage slightly increased from mid-April to a maximum at the beginning of May (28% (95% CI [22.8, 32.8]) mean coverage across all FPMs) and then slightly decreased until June 2018 (18% (95% CI [9.5, 25.9]) mean coverage) as adult amphibians migrated from the FPMs (Figure 1c). In 2019, the tadpole coverage followed the same pattern, with, however, a larger increase of tadpole populations until the beginning of May (42% (95% CI [27.7, 56.9]) mean coverage).

Differences in percentage tadpole coverages between individual ponds was higher in 2019 compared with 2018 (Figure 1c); FPM1, FPM5, FPM8 reached maximum coverages of around 80–90% in spring 2019, whereas FPM6, FPM7, and FPM10 had low coverage percentages of <20% (Table S7). Overall, tadpole percentage coverages were on average across all sampling dates highest in FPM1 and FPM5 and lowest in FPM6, FPM7, and FPM10 (Table S7).

3.1.4 | Submerged vegetation

The percentage areal coverage of submerged vegetation (i.e., algae and *Elodea spec.*) in the FPMs was low until the end of June 2018 (i.e., generally less than 20% coverage) and then substantially increased in July and August 2018 to around 50% coverage on average across all ponds (Figure 1d). After a phase of stabilization at around 50% coverage during autumn and winter 2018/2019, submerged vegetation coverage increased during spring and summer to generally >80% coverage in July 2019. A significant ($p < .001$) temporal trend for submerged vegetation coverage across all FPMs was indicated by GAM (Figure 1d).

Submerged vegetation coverage was comparable between the individual ponds until August 2018 (Figure 1d; Table S7). However, submerged vegetation development started to differ between the ponds from September 2018 onward, with FPM3 (the only FPM with coverage never exceeding 50%), FPM6, FPM7, and FPM10 continuously characterized by lower submerged vegetation coverages compared to the other ponds. In contrast, FPM2, FPM4, FPM5, FPM8, FPM11, FPM12 showed all above-average percentage submerged vegetation coverage from September 2018 onward (Figure 1d; Table S7). Apart from FPM3, algae and *Elodea spec.* vegetation coverage of the ponds aligned toward the end of the study period (Figure 1d).

3.1.5 | Terrestrial vegetation

During the first survey in November 2018, 53 grass and plant species were found at the 12 FPMs (see Table S8 for a full list of species).

Around 20 plant species were present at each individual FPM, with most species found at FPM2 (33 species) and FPM12 (28 species) (Figure 2a). Eight out of the 53 plant species were present at all FPMs (Table S8). During the second survey in May 2019, only four additional species (i.e., *Luzula spec.*, *Vicia sativa*, *Lotus corniculatus*, *Rumex crispus*) were detected.

3.1.6 | Ground beetles

Overall, 42 different ground beetle species and 1324 individuals were captured in the adjacent riparian floodplain areas and surrounding banks of the FPMs. The lowest number of species was detected in September/October 2018 (mean number of species across ponds: 2.3; 95% CI [1.7, 2.8]), while the highest number of species was captured in May 2019 (mean number of species across ponds: 10.9; 95% CI [9.7, 12.2]) (Figure 2b).

FPM12 had the overall highest species richness ($n = 18$), as well as the highest richness of one individual sampling date ($n = 17$) (Figure 2b; Table S9); in contrast, only 11 species were detected overall at FPM6. The number of individuals caught during the three sampling campaigns ranged from 71 (FPM2) to 133 (FPM1) (Table S9).

3.1.7 | Leafhoppers

Overall, 19 leafhopper species and 910 individuals were sampled in 2018 around FPM3, FPM5, FPM8, and FPM10 (please see Table S10 for a full list of species). The species numbers did not differ much between the four FPMs; however, the number of individuals was twice as high at FPM3 ($n = 326$) compared with FPM8 ($n = 163$) (Table S10).

3.2 | Aquatic and terrestrial litter decomposition

The microbially mediated leaf litter decomposition showed seasonal patterns in all FPMs (Figure 3a). $k_{\text{microbial}}$ increased from May 2018 ($k_{\text{microbial}}$ mean across all FPMs: 0.023; 95% CI [0.021, 0.026]) to an overall maximum in July 2018 ($k_{\text{microbial}}$ mean: 0.057; 95% CI [0.053, 0.062]), followed by a decrease to minimum decomposition values in November 2018 ($k_{\text{microbial}}$ mean: 0.011; 95% CI [0.009, 0.012]) and January 2019 ($k_{\text{microbial}}$ mean: 0.013; 95% CI [0.011, 0.014]). Subsequently, $k_{\text{microbial}}$ slightly increased again until June 2019 ($k_{\text{microbial}}$ mean: 0.021; 95% CI [0.019, 0.023]) (Figure 3a). The shredder-mediated leaf litter decomposition differed from the microbial decomposition as it (i) had consistently lower values particularly in summer 2018 and (ii) was rather constant over time, that is, the mean k_{shredder} was generally between 0.01 and 0.015 (Figure 3b). GAM results indicate significant overall temporal trends across all FPMs for both decomposition rates ($p < .001$; Figure 3a,b). We found rather small differences (i.e., a factor of 1.1) between $k_{\text{microbial}}$ and k_{shredder} during November 2018 and January 2019 (Figure 3a,b). Both decomposition rates were higher in summer 2018 than in summer 2019.

Considering the entire study period, average decomposition rates of the individual FPMs were comparable, with highest average values for $k_{\text{microbial}}$ (0.035; FPM1) and k_{shredder} (0.014; FPM3) not differing largely from lowest average values for both $k_{\text{microbial}}$ (0.026; FPM10) and k_{shredder} (0.01; FPM11) (Table S11). Nevertheless, larger differences in decomposition rates of the different FPMs existed particularly during the spring and summer months; differences between the ponds were moreover larger in 2018 and less pronounced in 2019 for both $k_{\text{microbial}}$ and k_{shredder} (Figure 3a,b).

Concerning terrestrial litter decomposition, weight losses per day were higher for green tea compared with rooibos tea. However,

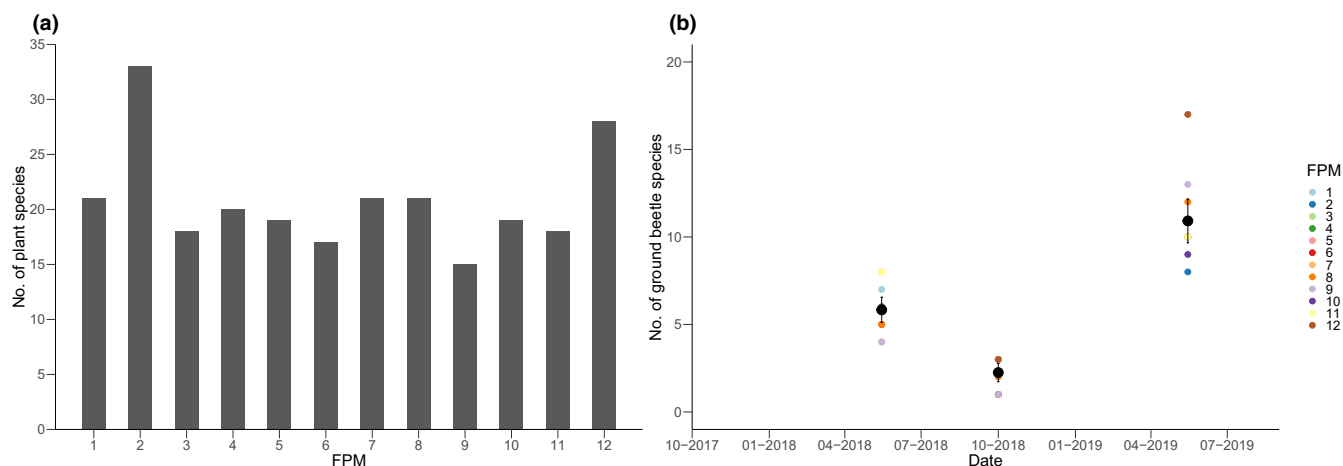


FIGURE 2 (a) Number of terrestrial plant species identified in November 2018 at the banks and floodplain areas of FPM 1–12, and (b) total number of ground beetle species sampled at the banks and floodplain areas of FPM 1–12 in May 2018, September/October 2018 and May 2019. Black circles in (b) depict means and error bars show 95% confidence intervals

the terrestrial litter decomposition for all ponds showed a comparable pattern over time for the two different tea bag types, with significant ($p < .001$) overall temporal trends for both parameters

identified by GAM (Figure 3c,d). Litter decomposition increased during spring and peaked in summer and early autumn 2018 (highest mean decomposition rates across all FPMs: 48.27 mg/day (95% CI

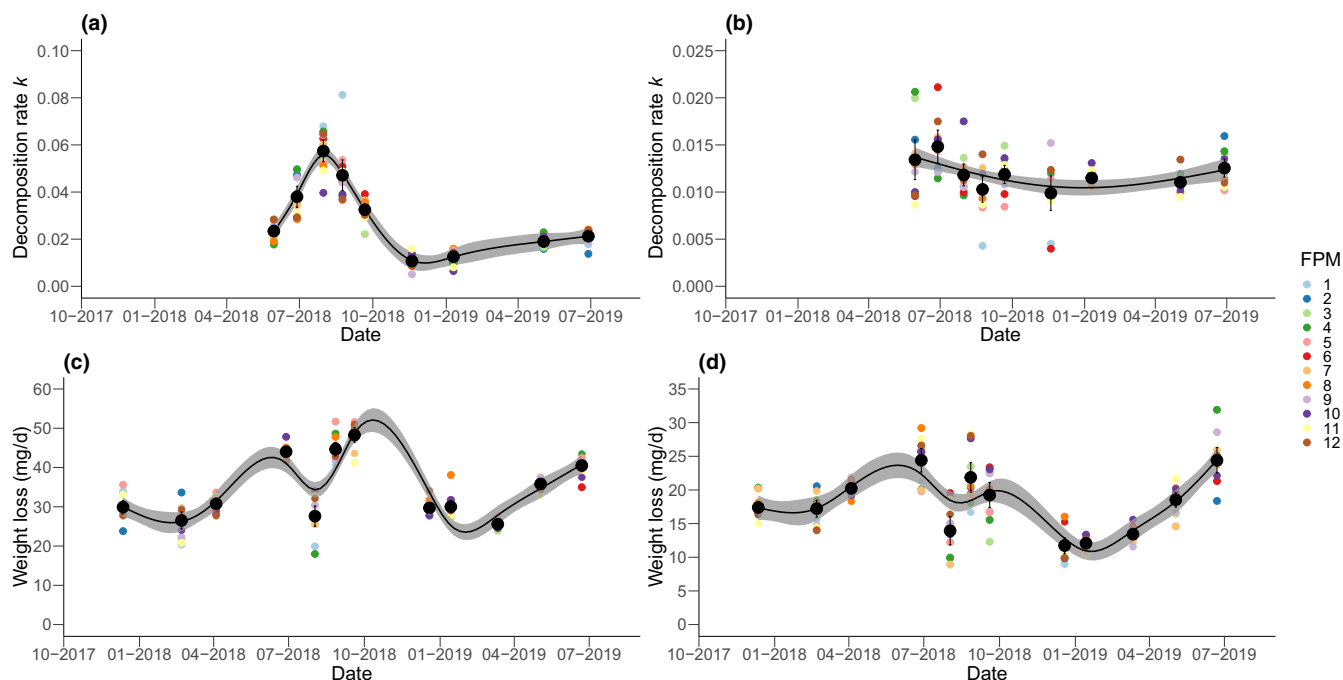


FIGURE 3 Generalized additive model (GAM) plots for leaf litter decomposition expressed as decomposition rate k for (a) microbial decomposition ($k_{\text{microbial}}$) and (b) shredder-mediated decomposition (k_{shredder}) over time. GAM plots for the terrestrial litter decomposition at the banks of the twelve FPMs expressed as linear weight loss per day (mg/day) for (c) green tea and (d) rooibos tea over time. All values in (a–d) are displayed at the end of each sampling period. Black circles depict means and error bars show 95% confidence intervals. The gray shaded areas in (a–d) indicate the 95% confidence intervals

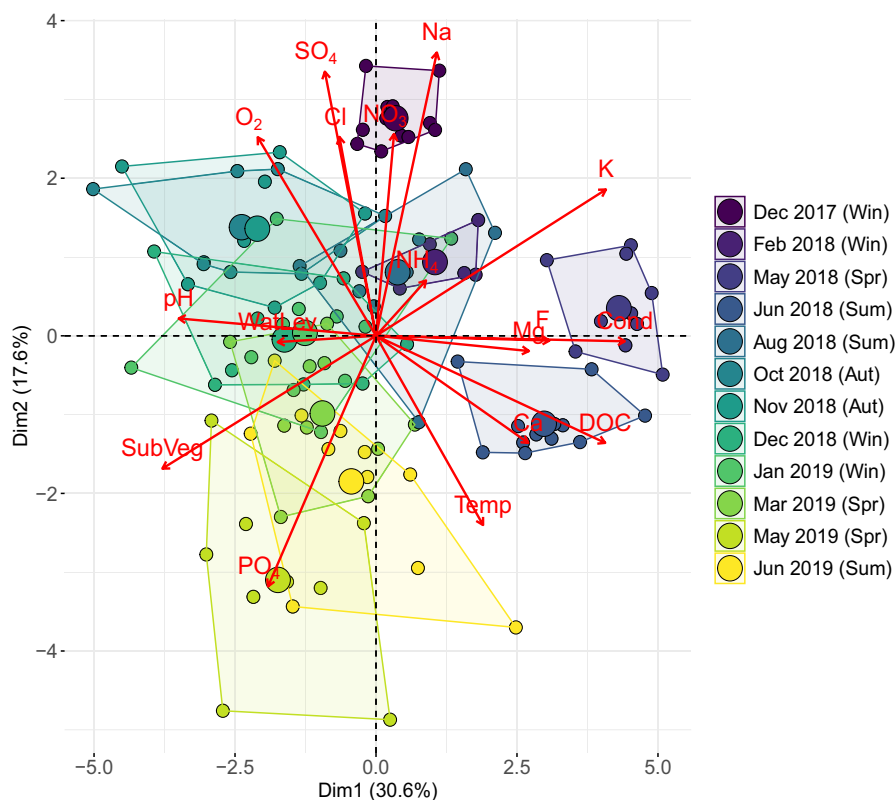


FIGURE 4 PCA biplot of ecosystem development and environmental conditions in the twelve FPMs for the entire study period. Small color-coded (see figure legend) dots and polygons represent FPM observations within months (with respective seasons) across the study; larger dots represent the mean for a given month

TABLE 1 VPA results for the factors successional time, season, and pond for the entire study period, as well as for the first and second year of the study

Factor	Explained variance (entire study period)	Explained variance (1 st year)	Explained variance (2 nd year)
Pond	0.062	0.072	0.335
Season	0.045	0.226	0.150
Successional time	0.361	0.087	0.013
Interaction successional time—seasons	0.148	0.387	0.208

[46.43, 50.12]) for green tea (September 2018) and 24.41 mg/day (95% CI [22.66, 26.16]) for rooibos tea (June 2018)) and decreased in late autumn and during winter months (lowest mean decomposition rates across all FPMs: 25.57 mg/day (95% CI [24.90, 26.24]) for green tea (March 2019) and 11.74 mg/day (95% CI [10.58, 12.89]) for rooibos tea (December 2018)). The weight loss in tea bags was high already at the first sampling date in December 2017 and did not increase when comparing subsequent seasons (Figure 3c,d).

Terrestrial litter decomposition showed generally largest inter-pond variations during summer and early autumn months, but were overall comparable across the study period (Figure 3c,d; Table S12). Particularly for green tea, the interpond variation was higher in 2018 compared with 2019.

3.3 | Statistical evaluation of pond ecosystem developments over time

Two PCA axes were selected by the broken stick approach, explaining a total variance of 48.2%. The first PCA axis, which explains 30.6% of the total variance, has significant loadings (correlation > .60) on specific conductivity, potassium, DOC, fluoride, as well as submerged vegetation and pH. The second PCA axis explains 17.6% of the total variance and correlates to sodium and sulfate, as well as to phosphate (Table S13).

For the entire study period, no linear pattern of pond environmental condition was identifiable along either the first or second PCs (Figure 4). However, the PCA biplot indicates differences in ordination patterns between the first year (i.e., winter 2017/2018 until summer 2018) and the second year (i.e., winter 2018/2019 until summer 2019) of pond ecosystem development (Figure 4); overall, there were higher differences in pond environmental conditions in the first year compared to the second year. The PCA indicates that the differences in environmental conditions between the different seasons (i.e., winter, spring, summer) are larger in the first year compared with the second year while the interpond diversity in the respective months/seasons was higher in the second year and largest at the end of the study period (i.e., spring and summer 2019). The PCAs performed separately for the first and second year confirmed these results. In the first year, large differences in environmental conditions of the ponds existed among the different seasons (i.e., large seasonal effects), driven by, for example, water temperature and dissolved oxygen (Figure S9a); the interpond variability within each season was, however, small, particularly in winter 2017/2018 and spring 2018. In contrast, the PCA

biplot for the second year shows less effects of the seasons in driving overall differences in environmental conditions, while environmental differences among ponds appear to become more influential (Figure S9b). In the same PCA ordination space, when the pond is selected as grouping factor, overall no large differences in environmental conditions are visible among ponds if the entire study period is considered (Figure S10a). Also in this case, when the PCA is constrained to the first and second years, a high similarity of pond environmental conditions is shown until summer 2018 (Figure S10b), whereas differences in environmental conditions among ponds increase substantially in the second year (Figure S10c). This also confirms a lower interpond environmental diversity within seasons (i.e., less seasonal effects) in the second compared to the first year. A further PCA shows that the pond location (i.e., left, middle, right) along the experimental site (see Figure S1) had no effect on environmental conditions of the ponds (data not shown).

The results of the RDA and VPA overall confirm patterns observed in the PCA ordination. RDA and VPA show that for the entire study period successional time (i.e., time since the beginning of the study, VPA explained variance = 36.1%) and its combined effect with seasonality (VPA explained variance = 14.8%) have the largest effect on the environmental conditions of the FPMs (Table 1; Table S14). Similarly to what was observed in the PCA, when the RDA is performed exclusively on the first year of ecosystem development, it identifies season (VPA explained variance = 22.6%) and its interaction with successional time (VPA explained variance = 38.7%) to explain most variance in the environmental conditions of the FPMs (Table 1), whereas the factor pond had no significant influence (Table S15). On the other hand, when the RDA is performed on the second year, analysis identifies pond as significant factor explaining the highest proportion of variance (VPA explained variance = 33.5%) in environmental conditions, followed by the interaction of seasons with successional time (20.8%; Table 1; Table S16).

4 | DISCUSSION

4.1 | Pond ecosystem developments and environmental conditions

Monitoring of the first 21 months of initial pond development and primary succession revealed dynamic ecosystem successional trajectories in the twelve FPMs (see also Figure S2 and the time-lapse video available at: <https://youtu.be/vfTNpwyYhFE>). Data from the

present survey show that newly established ponds are colonized quickly within weeks or few months; the development of pond ecosystem structure and function is thereby shaped progressively by seasonal influence and the varying abiotic and biotic conditions. The overall initial ecosystem development from the “point zero” observed here for the FPMs is in line with pond studies under field conditions. Similar developmental patterns were, in fact, also observed at a larger scale for the Chicken creek catchment in eastern Germany (Hüttl et al., 2013). Also, our results are comparable to the findings from the Pinkhill Meadow pond complex in the UK, in which a rapid pond colonization and ecosystem development was observed within the first years after construction (National River Authority, 1992).

Fluctuations of physico-chemical water quality parameters over time (Appendix S1, supporting results) likely resulted from effects caused by seasonality (effects on e.g., water temperature), meteorology (effects of precipitation on e.g., conductivity), and biology (e.g., effects of photosynthesis by aquatic vegetation on pH and dissolved oxygen concentrations). Overall low inorganic ion concentrations are related to the fact that ponds have been entirely filled with water from the Sulzbach, a nutrient- and ion-poor stream. Also, interpond variability of physico-chemical parameters could be explained by differences either in submerged vegetation coverage (pH values; dissolved oxygen) or differences in soil entries caused by bank erosion (e.g., specific conductivities; ion concentrations). Intersystem variability of physico-chemical parameters has also been observed in previous studies with replicated outdoor pond mesocosms (e.g., Caquet et al., 2001; Christman et al., 1994). However, care must be taken in interpreting the dissolved oxygen concentration data particularly of FPM9-12 as there are larger data gaps due to O₂ logger failures (i.e., defective devices).

PCA and RDA indicate environmental conditions during the successional development of the FPMs to be driven by a resulting effect of succession time, seasonality, and their interaction. The factor successional time was an important driver during the entire study period and particularly during the first year of monitoring, which can be expected for the development of newly established ecosystems after a “point zero” (Miguel-Chinchilla et al., 2014; Williams et al., 2008). The interaction of successional time with seasonality resulted in changing and nonlinear patterns over time. A large effect of seasonality on the pond environmental conditions during the first year, and environmental homogeneity among the twelve ponds (i.e., low interpond variability) suggests that external environmental cues (e.g., meteorological conditions in the different seasons) were the main drivers for this initial phase of ecosystem development, whereas internal ecological processes within the FPMs played a minor role, resulting in low interpond variation. The seasonal influence on pond environmental conditions decreased in the 2nd year, while the interpond diversity concurrently increased due to natural environmental divergence. External factors (e.g., season-driven meteorological effects) became thus less important over time, whereas internal factors and processes in the individual ponds led to more diverse ecosystem formations and significant differences in environmental condition and habitat structure; these developments led

to increasing complexities and heterogenization of the FPMs. Our findings, comparable to those found in the Chicken creek catchment study (Elmer et al., 2013; Hüttl et al., 2013), confirm that while external factors dominate the very initial phase of ecosystem development, internal interacting abiotic and biotic factors gained importance in later stages. Increases in interactions and feedback, accompanied by, among other, an increase in biomass and species diversity (see below), are clear indicators of ecosystem development and succession (Fath et al., 2004; Odum, 1969).

4.2 | Pond colonization and aquatic and terrestrial community development

Data from our study show that new, undeveloped ponds are colonized quickly, that is, zooplankton and invertebrate species were recorded in six ponds already at the initial sampling dates (15 November 2017 and 11 December 2017); algae and *Elodea* spec. were found in all ponds during the first survey in January 2018. A fast colonization by aquatic invertebrates and aquatic vegetation was also observed within the first year after construction in four ponds of the Pinkhill Meadow pond complex (National River Authority, 1992; Williams et al., 2008) and in newly created ponds in Wales (Gee et al., 1997). The inherent mobility of freshwater taxa fosters their dispersal and colonization of new habitats (Williams et al., 2008); existing (semi-) natural ponds and small streams at the EERES site within a distance of less than 100 m from the FPMs likely acted as source habitats for plant and animal propagules colonizing the twelve FPMs. In addition, filling the ponds with water from the Sulzbach most likely contributed to their colonization by actively introducing different taxa such as benthic larvae. At the same time, new pond habitats are characterized by specific environmental conditions, that is, they are dominated by inorganic substrates, have a low coverage of aquatic macrophytes, and lack top predators, such as fish (Williams et al., 2008). Colonizing species and taxa with short generation times such as zooplankton, invertebrates, and algae benefit from such initial conditions and are able to establish populations as observed here. Diverse terrestrial plant communities, as well as leafhopper and ground beetle populations, rapidly colonized the terrestrial component of the FPMs within a few months after the ponds' establishment with overall more than 100 species. Such a rapid colonization of the terrestrial component of the ponds is in line with colonization patterns of terrestrial plants observed at the Pinkhill pond complex site (Williams et al., 2008). Surveys of the terrestrial communities were not conducted as regularly as the surveys of the aquatic components and did not start directly after the establishment of the ponds. For this reason, drawing conclusions about the very initial colonization phase of the terrestrial banks and floodplain areas of the FPMs is limited.

However, the overall increase of aquatic and terrestrial taxon richness and abundances in all FPMs over the seasons (i.e., higher richness and abundances in subsequent seasons compared to previous seasons) indicates progressive ecosystem development and

succession over time. Overall, the colonization and developments of the aquatic and terrestrial communities were comparable between the different ponds, but differences existed in terms of species presence and development of population sizes for multiple aquatic and terrestrial groups, individual sampling dates and across the entire study period. For instance, overall numbers of macroinvertebrate abundances or families of emerged merolimnic insects detected during the study period differed by 450% and 400%, respectively, for individual ponds. The observed increase in variation of the aquatic and terrestrial communities of the FPMs over time also evidences ecosystem development and succession as maturing systems become more structured and less homogenous than systems at earlier stages (Elmer et al., 2013; Fath et al., 2004); the increased biological divergence is in line with the heterogeneity observed in PCA, RDA, and VPA for the environmental conditions (see above), with the physical location of the FPMs not explaining taxa richness and abundances of the ponds. Generally, biological interactions in initial ecosystems are limited so that it is likely that, during the initial phase of development, aquatic communities in the FPMs were shaped by (i) bottom-up effects induced by habitat quality and the physico-chemical environments, or (ii) stochastic processes determining species pond colonization. Given that the FPMs are located in close proximity, are similar in size, depth, substrate type, and climatic conditions, and have comparable physico-chemical characteristics particularly during the first year (see discussion above), differences observed among ponds for taxa richness and abundances during initial ecosystem developments are unlikely to be explained by abiotic environmental conditions (although this relation was not statistically analyzed in this study). Thus, stochastic effects of early FPM species colonization and propagating community effects are more likely explanations for differences in aquatic communities and abundances observed between the FPMs. However, interpond taxa variability (i.e., differences in species numbers and abundances among the FPMs) increased over time, alongside those of abiotic parameters. This likely is a result of ecosystem divergence (see discussion of PCA and RDA results above), priority effects after the stochastic initial phase (Weidlich et al., 2021), as well as increasing biotic-abiotic interactions and feedback in the ponds during their advancing ecosystem development. Street and Titmus (1979) showed for macroinvertebrate communities of six newly constructed experimental ponds that interpond taxa variability was driven by increasing variability of pond environmental conditions and habitat structure (e.g., differences in aquatic macrophyte abundances) within two years of observation. Also, the importance of stochastic sequences of species arrival and subsequent priority effects, in which the initial species assemblages partly determines future community developments, has been largely proven for the colonization of new pond habitats (Chase, 2003; Louette & De Meester, 2007); priority effect-induced long-lasting differences in species dominance and overall community assemblages may thus lead to future increases in taxa variability among FPMs. Long-term pond monitoring in the UK showed larger variations of plant richness and macroinvertebrate richness in ponds after seven years compared to the first year of

construction (Williams et al., 2008). Overall, as the FPM communities mature, individual ecosystem properties including abiotic and biotic interactions are expected to increase further in influence over time (Frisch et al., 2012), potentially resulting in further increase of interpond taxa variability.

4.3 | Pond ecosystem development and ecosystem functioning

Decomposition of plant litter denotes an important ecosystem function for the provisioning of energy and nutrients in aquatic and terrestrial environments (Benfield et al., 2017; Krishna & Mohan, 2017). However, concerning leaf litter decomposition in the FPMs, data were not available before May 2018, so that the aquatic decomposition performances during the very initial stages of ecosystem development could not be determined. Available data, however, show that the leaf decomposition in the FPMs was mostly driven by microbial decomposition processes and less by invertebrate shredder-mediated decomposition, particularly in summer 2018. This can be explained by the fact that microorganisms (i.e., aquatic fungi and bacteria) have very short generation times, which enables faster colonization and microbiological community growth in the initial pond ecosystems. Moreover, the data indicate that $k_{\text{microbial}}$ is strongly determined by seasonal effects, that is, leaf litter decomposition is higher during periods with higher water temperatures and lower during colder months. This was expected given that higher water temperatures foster the abundance and activity of microorganisms and thus $k_{\text{microbial}}$ (Martinez et al., 2014). However, in comparison to $k_{\text{microbial}}$, k_{shredder} was substantially less influenced by seasonal effects, which is possibly explained by the year-round presence of macroinvertebrates in the FPMs.

In contrast to the increase of numbers of aquatic species and abundances in subsequent seasons, a season-specific comparison shows that the ecosystem function leaf decomposition decreased over time, which is most pronounced for $k_{\text{microbial}}$; slightly lower water temperatures in spring and summer 2019 compared to 2018 (Appendix S1, supporting results) might be an explanation. Interestingly, in contrast to what we found for all other abiotic and biotic variables (see above), the differences in leaf litter decomposition rates between the ponds also decreased over time. It follows that this ecosystem function converged with progressing ecosystem developments of the FPMs. One reason for the taxonomical divergence but functional convergence observed here for the second year might be that later arriving species are not yet able to fully compete with species already established in the FPMs after a largely stochastic initial colonization and niches (i.e., energy sources such as litter decomposition) partitioned already between the initial colonizers.

Data from the terrestrial parts of the FPMs in December 2017 revealed terrestrial litter to be efficiently decomposed already under initial ecosystem conditions. Moreover, terrestrial litter decomposition did not increase over time when comparing the respective seasons. This is in line with our findings on aquatic leaf

litter decomposition and once again in contrast to our taxonomical findings on the structural ecosystem development, which showed a season-related increase in aquatic and terrestrial species presence and abundances between the FPMs. That said, ecosystem functions such as aquatic and terrestrial litter breakdown are, however, effective already in the very initial stages of freshwater ecosystem development, whereas ecosystem structure (i.e., diversity and abundance of different taxa) gradually increases over time. However, terrestrial litter decomposition was also strongly influenced by seasonal and meteorological effects, with higher soil temperature and soil moisture fostering an increase in litter decomposition due to elevated microbial activity in warmer months (Krishna & Mohan, 2017). Regarding the latter, the prolonged dry period in July 2018 (Appendix S1, supporting results) may explain the substantial reduction in litter degradation at the beginning of August 2018. The faster decomposition of green tea compared with rooibos tea is well known from the literature and explained by differences in litter composition, particularly cellulose and lignin contents (Didion et al., 2006; Keuskamp et al., 2013).

5 | CONCLUSION AND OUTLOOK

In our study, the initial freshwater ecosystem development and primary succession could successfully be observed at the EERES site. Using an array of twelve FPMs denotes a novel and promising approach to observe and disentangle abiotic and biotic conditions including structural and functional ecosystem developments as well as their complex interactions. The FPMs at the EERES site offer unique research opportunities to study small freshwater environments, ecosystem development, and water-land interaction under field conditions in a replicated and controlled manner. The ecological research possibilities at the EERES site thus differ from other ecosystem research approaches that either analyze mature ecosystems, or do not feature repeated designs or sites with clear boundary conditions and fully controllable research environments. The findings on ecosystem primary succession and initial ecosystem developments from a "point zero" presented here may benefit planning, implementation, and ecological evaluation of freshwater restoration measurements, which are widely implemented in Germany and Europe (see e.g., <http://www.europeanponds.org>; <https://www.ecrr.org>). Also, our monitoring data provide valuable information for the ecological assessment of temporary ponds with seasonal formation in river floodplain areas. Moreover, consequences from anthropogenic disturbances of natural pond ecosystems may become more predictable if fundamental mechanisms of initial ecosystem colonization are better understood. Our findings also support that creating small ponds denotes a simple and cost-effective tool to enhance landscapes by biodiversity-rich habitats within short periods of time; this is of particular importance considering the ongoing decline of global freshwater biodiversity (e.g., Albert et al., 2021; Tickner et al., 2020). Last, our study shows that particularly interpond taxa variability among replicated outdoor pond mesocosms in early successional

stages must be considered in case these aquatic systems are used to analyze responses to anthropogenic stressors, such as chemicals (Caquet et al., 2000, 2001). Although larger mesocosms such as the FPMs used here are more internally stable and sustainable, which reduces intersystem variability compared with smaller systems (Belanger, 1997; Caquet et al., 2000), larger experimental pond systems also need to be managed (e.g., inserting species into ponds; mixing water and sediments between ponds) to increase replicability between the experimental units.

Future research conducted at the EERES FPMs can range from ecological studies without any anthropogenic manipulation up to entirely controlled and manipulated studies with, for example, highly fluctuating water levels and flooding regimes. The EERES research site is open to future research cooperations in all kind of aquatic and terrestrial research projects. The twelve FPMs of the EERES site are currently used for experimental research of the DFG Research Training Group 2360 "SystemLink" (<https://systemlink.uni-landau.de>), which investigates biogeochemical and ecological interactions of aquatic and terrestrial ecosystems under anthropogenic stress.

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CONFLICT OF INTEREST

The authors have no conflict of interest.

AUTHOR CONTRIBUTIONS

Sebastian Stehle: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Validation (lead); Visualization (equal); Writing – original draft (lead); Writing – review & editing (equal). **Alessandro Manfrin:** Formal analysis (lead); Methodology (equal); Writing – review & editing (equal). **Alexander Feckler:** Visualization (equal); Writing – review & editing (equal). **Tobias Graf:** Data curation (equal); Investigation (equal). **Tanja J. Joschko:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Methodology (equal); Project administration (lead). **Jonathan Jupke:** Formal analysis (equal); Methodology (equal); Writing – review & editing (equal). **Christian Noss:** Conceptualization (equal); Investigation (equal); Writing – review & editing (equal). **Verena Rösch:** Investigation (equal); Writing – review & editing (equal). **Jens Schirmel:** Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – review &

editing (equal). **Thomas Schmidt:** Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Jochen P. Zubrod:** Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Ralf Schulz:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY STATEMENT

All data of this study are provided in the main text and in the appendix.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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Appendix S1

Supporting Methods



Figure S1. Aerial image of the twelve floodplain pond mesocosms (FPM 1-12) at the Eußerthal Ecosystem Research Station (EERES). Aerial photo taken in April 2018.

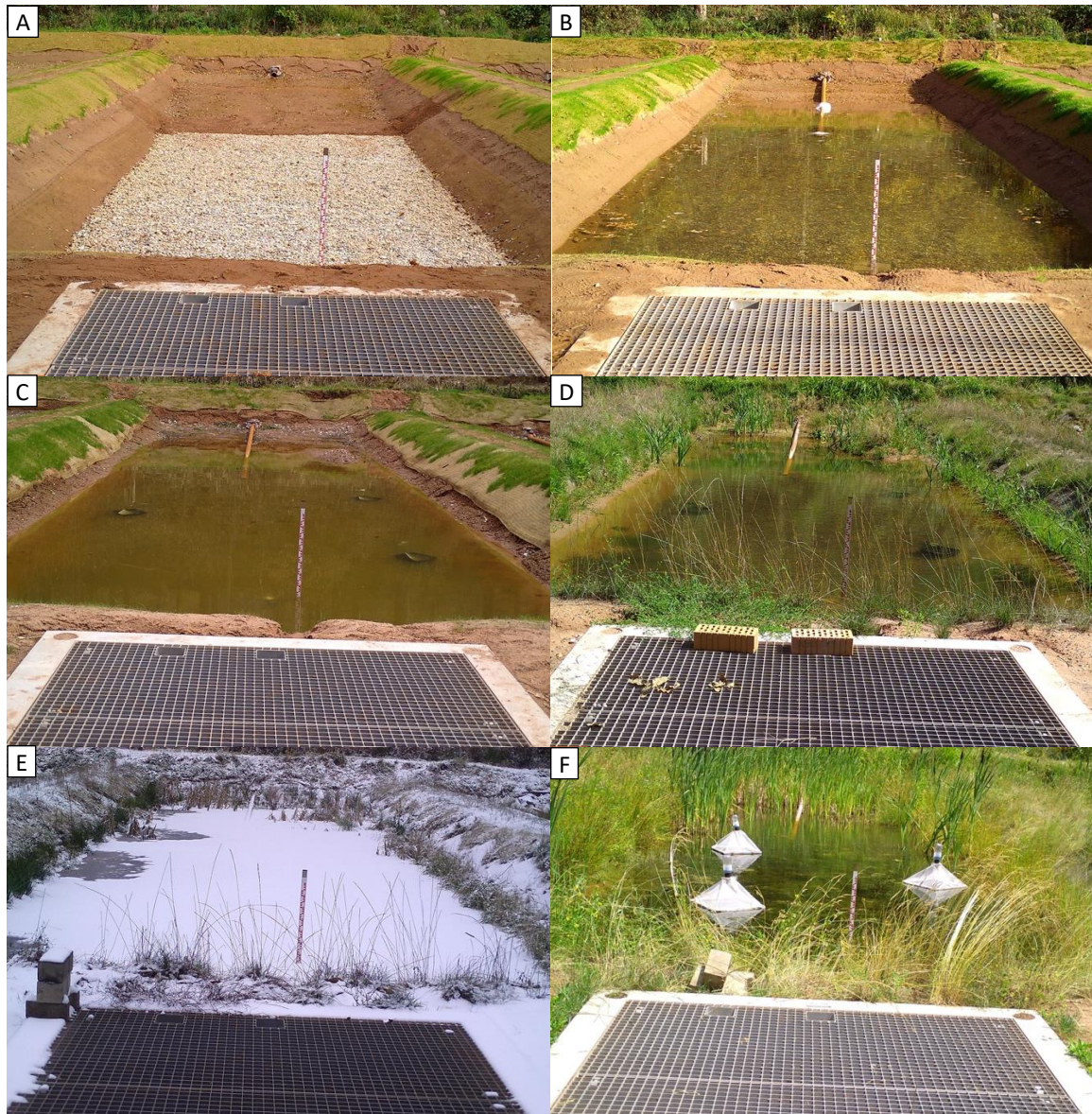


Figure S2. Exemplary photos of the ecosystem development of FPM10 from October 2017 till July 2019. Photos showing FPM10 in (A) October 2017, (B) November 2017, (C) March 2018, (D) August 2018, (E) January 2019, (F) July 2019. Please see also <https://youtu.be/vfTNpwyYhFE> for a time lapse video of the ecosystem development during the study period.

Table S1. Overview of the monitoring program, sampling locations, sampling methods and sampling intervals.

Parameter	Sampling location [†]	Method (sampling interval)
General information on EERES site and FPM conditions		
Ecosystem development of the floodplain mesocosms	FPM 1-12, banks and water-land floodplain area	Manual photos (weekly); automatic photos by camera trap (twice daily)
Weather data	Eußerthal / Siebeldingen	Meteorological station (Hourly)
Soil temperature (5 cm depth)	North and south banks of FPM 1-12	Digital probe thermometer (monthly)
Water level	FPM 1 -12	Level gauge (monthly)
Physico-chemical water quality parameters		
Water chemistry (DOC, TOC, anions, cations)	FPM 1-12	Analytical measurement (monthly – bimonthly)
Specific conductivity	FPM 1-12	Portable analytical device (monthly)
pH	FPM 1-12	Portable analytical device (monthly)
Water temperature	FPM 1-12	Temperature logger (10 minutes)
Dissolved oxygen	FPM 1-12	O ₂ logger (10 minutes)
Biotic parameters		
Benthic macroinvertebrates	FPM 1-12	Three pebble baskets per FPM (monthly – bimonthly)
Merolimnic insect emergence	FPM 1-12	Three emergence tents per FPM (weekly)
Zooplankton	FPM 1-12	Plankton net (monthly)
Amphibians (tadpoles)	FPM 1-12	Visual inspection of percentage tadpole coverage (daily – weekly while present)
Submerged vegetation	FPM 1-12	Visual inspection of percentage filamentous algae and <i>Elodea</i> coverage (14 d – monthly)
Crayfish and fishes	FPM 1-12	Three crayfish traps per FPM (monthly)
Terrestrial vegetation	Banks and water-land floodplain areas of FPM 1-12	Plant mapping (yearly)
Ground beetles	Banks and water-land floodplain areas of FPM 1-12	Four pitfall traps (twice in 2018 and once in 2019)
Leafhoppers	Banks of FPM3, FPM5, FPM8, FPM10	Suction sampling (June and September 2018)
Functional parameters		
Aquatic leaf litter decomposition	FPM 1-12	Three coarse and fine leaf bags per FPM (monthly)
Terrestrial litter decomposition	North and south banks of FPM 1-12	Two tea bags (rooibos and green tea) (monthly – every three months)

[†]: If not specified otherwise, FPM 1-12 indicates that the sampling took place in the water body of all twelve FPMs.

Supporting Results

Meteorology

The meteorological data for the study period were derived from two weather stations in the vicinity (8 km and 1 km distance) to the EERES site. Rainfall data (mm per hour and per day) were taken from the meteorological station Eußerthal (<https://bit.ly/31XVF75>) and air temperature (°C, hourly and daily mean), as well as global radiation (Wh/m², sum per day and month) from the meteorological station Siebeldingen (<https://bit.ly/2OxOTI9>). The meteorological data for the study period (06 November 2017 – 31 July 2019) are provided in Figure S3 and Table S2. The air temperature had minima in February 2018 (hourly minimum of -10.6 °C) and January 2019 (hourly minimum of -7.8 °C), whereas maximum temperatures were recorded at the end of July and beginning of August 2018 (hourly maximum of 34.3 °C), as well as in June and July 2019 (hourly maximum 36.8 °C) (Figure S3A; Table S2). The daily sum of global radiation shows minima below 100 Wh/m² in the winter month of 2017/2018 and 2018/2019, whereas maxima of > 8000 Wh/m² occurred in June and July of 2018 and 2019 (Figure S3B); highest sum global radiations per month were recorded for July 2018 and July 2019 (Table S2). The precipitation data revealed particularly dry periods in July and October 2018, with precipitation sums of less than 20 mm per month (Figure S3C; Table S2). In contrast, high precipitation amounts occurred at the beginning of the study period (i.e., November 2017 – January 2018), as well as in May 2018, in which also the highest amount of rainfall per hour (40.23 mm) occurred (Figure S3C; Table S2).

Soil temperature

The soil temperature in 5 cm depth was measured monthly on the top of the northern and southern banks of each FPM using a digital probe thermometer (TFA Dostmann). The soil temperature followed a seasonal pattern with lowest temperatures in the winter months and highest during summer (Figure S4). The soil temperatures were generally higher at the northern banks due to higher solar irradiation reasoned by their southward direction (Figure S4; Table S3). Differences in soil temperature between individual ponds occurred particularly during the summer months (Figure S4); these differences are, however, likely the results of measurement bias, i.e., the temperature was in parts measured at different times of the day at the individual ponds. However, the median values of the soil temperature of both the northern and southern banks were comparable indicating no general differences between FPMs (Table S3).

Water level

The water levels of the twelve FPMs were recorded monthly from a level gauge at the outlet site of the ponds. Overall, the levels were comparable across the twelve ponds, with median water levels of 30 cm documented for all FPMs (Table S4). Individual outlier values of 20 cm (FPM11) and 38 cm (FPM1) in February and March 2018 are related to pond icing. However,

no general increasing or decreasing trend exists for the water levels over the study period (Figure S5).

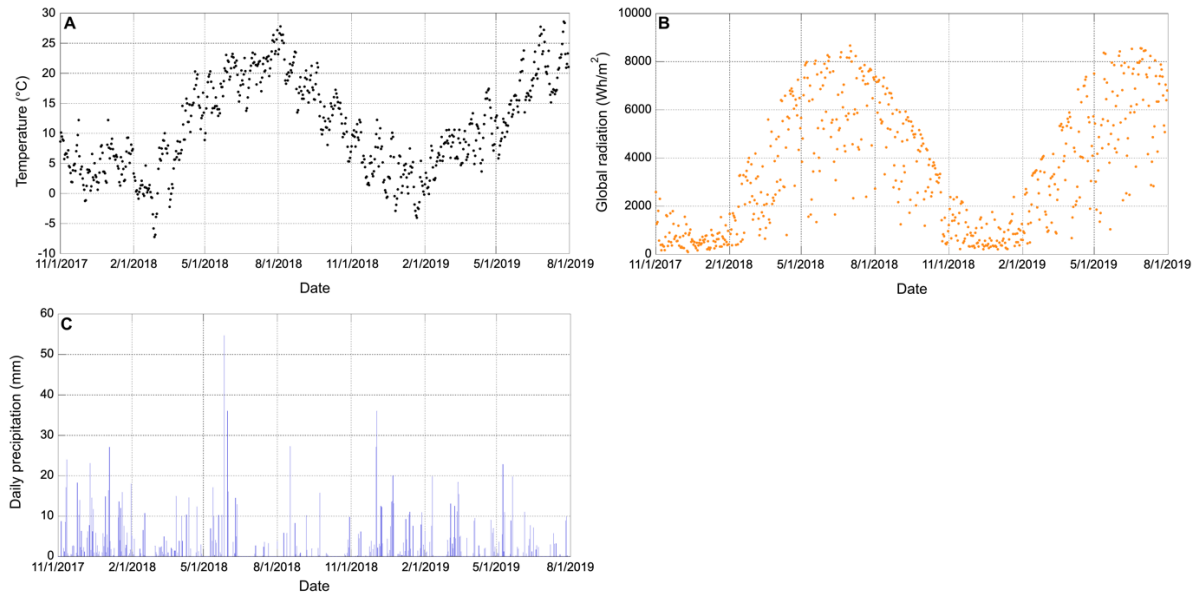


Figure S3. Meteorological data for the EERES site, with (A) air temperature in °C (daily mean), (B) global radiation in Wh/m² (daily sum per day) and (C) precipitation in mm (sum per day) displayed for the study period 06 November 2017 to 31 July 2019.

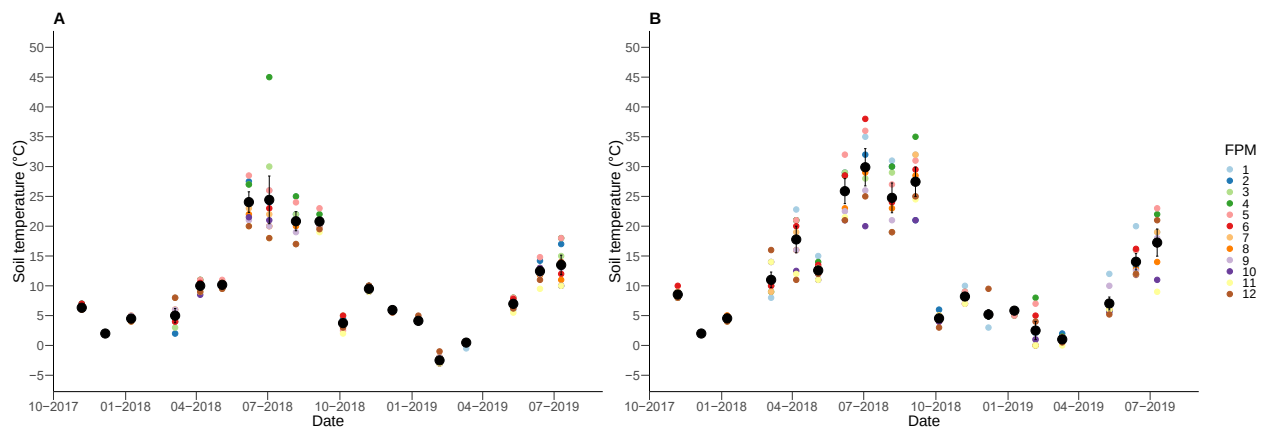


Figure S4. Soil temperature in 5 cm depth of (A) southern and (B) northern banks of the twelve FPMs. Black circles depict means and error bars show 95% confidence intervals.

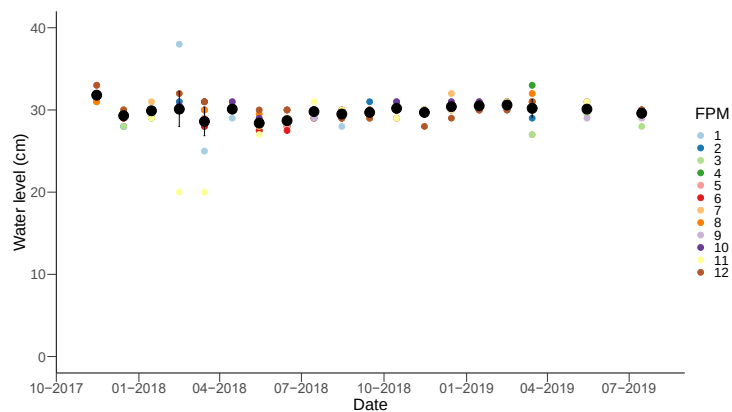


Figure S5. Water levels of FPM 1-12 recorded from level gauges during the study period. Black circles depict means and error bars show 95% confidence intervals.

Table S2. Monthly characteristics of precipitation amounts, air temperature and global radiation for the EERES site.

Date	Precipitation (mm) per hour					Air temperature (°C)				Global radiation (Wh/m ²)
	Mean	Median	Max.	Min.	Sum per month	Mean	Median	Max.	Min.	Sum per month
Nov 17	0.17	0.00	6.78	0.00	123.6	5.90	5.70	17.30	-1.00	27892
Dec 17	0.17	0.00	4.00	0.00	123.1	3.53	2.90	14.10	-2.50	17273
Jan 18	0.21	0.00	7.06	0.00	154.7	5.87	6.20	12.10	-2.80	19122
Feb 18	0.05	0.00	2.19	0.00	31.5	-0.05	0.50	9.50	-10.60	57171
Mar 18	0.07	0.00	3.28	0.00	48.9	4.68	5.55	13.50	-7.50	77496
Apr 18	0.09	0.00	9.61	0.00	63.1	13.83	13.30	27.10	-0.30	154311
May 18	0.21	0.00	40.23	0.00	158.4	16.83	16.40	29.10	2.20	177209
Jun 18	0.07	0.00	14.45	0.00	52.9	19.55	19.10	30.10	7.60	183302
Jul 18	0.02	0.00	2.00	0.00	12.9	22.16	21.25	34.30	12.90	202881
Aug 18	0.08	0.00	23.06	0.00	55.8	20.95	20.20	34.20	6.80	168742
Sep 18	0.04	0.00	11.70	0.00	31.8	16.35	16.10	30.40	2.60	130776
Oct 18	0.03	0.00	2.51	0.00	19.6	11.93	11.50	25.70	0.70	77223
Nov 18	0.04	0.00	2.93	0.00	29.6	6.60	5.90	16.70	-2.40	37330
Dec 18	0.24	0.00	5.39	0.00	175.1	4.54	5.00	13.10	-4.50	20264
Jan 19	0.11	0.00	2.60	0.00	80.1	1.70	1.90	8.80	-7.80	24924
Feb 19	0.06	0.00	3.64	0.00	40.1	5.11	4.50	18.00	-3.40	69869
Mar 19	0.15	0.00	3.77	0.00	110.8	8.16	8.15	19.90	-1.20	104757
Apr 19	0.07	0.00	7.12	0.00	52.0	10.73	10.10	23.80	-0.20	133114
May 19	0.11	0.00	5.67	0.00	79.6	12.53	12.30	23.60	1.60	160382
Jun 19	0.06	0.00	7.42	0.00	42.7	20.42	19.70	36.50	8.30	196863
Jul 19	0.05	0.00	8.41	0.00	34.5	20.91	20.20	36.80	8.30	199729

Table S3. Statistical characteristics of the soil temperature (°C) at 5 cm depth of the southern and northern FPM banks.

South bank	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	-3.0	-3.0	-3.0	-2.0	-3.0	-2.0	-2.0	-3.0	-2.0	-3.0	-3.0	-1.0
25th percentile	3.8	4.5	4.3	4.0	4.0	4.3	4.3	4.3	4.5	4.3	4.0	4.5
Median	6.0	7.0	8.0	8.0	8.0	7.8	6.8	6.8	6.2	8.0	8.0	8.0
75th percentile	11.8	15.6	14.0	15.0	16.4	12.1	13.5	11.6	12.5	11.5	9.8	12.5
90th percentile	25.2	22.4	23.0	25.4	24.4	21.4	21.2	20.0	20.0	21.0	18.2	18.3
Maximum	27.0	27.5	30.0	45.0	28.5	24.0	23.0	22.0	21.0	21.5	20.0	20.0
Mean	9.6	10.1	10.3	11.5	10.6	9.6	9.5	9.0	9.0	8.9	8.2	8.8
North bank	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	1.0	0.0	1.0	1.5	1.1	1.3	1.0	0.0	0.5	0.8	0.0	0.5
25th percentile	5.3	5.4	4.7	5.0	5.0	5.0	5.0	5.0	4.8	4.8	5.0	4.6
Median	10.0	9.0	9.0	8.3	9.0	10.0	8.5	8.5	10.0	8.0	7.5	9.5
75th percentile	21.4	19.0	19.0	19.8	22.0	18.5	17.5	15.0	17.0	13.3	12.0	17.5
90th percentile	31.2	29.2	28.2	29.0	31.2	28.7	25.3	24.1	23.0	20.2	19.8	21.8
Maximum	35.0	32.0	29.0	35.0	36.0	38.0	32.0	29.0	26.0	21.0	24.5	25.0
Mean	14.0	12.3	12.3	12.7	13.8	13.1	11.3	11.4	11.2	9.8	9.2	11.2

Table S4. Statistical characteristics of the water levels of the FPMs during study period (06 November 2017 – 31 July 2019).

Water level (cm)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	25.0	28.0	27.0	27.5	28.0	27.5	28.0	29.5	28.5	29.0	20.0	28.0
25th percentile	29.0	29.5	28.8	29.0	30.0	29.0	30.0	30.0	29.0	30.0	29.0	30.0
Median	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0
75th percentile	30.0	30.3	30.0	30.0	31.0	30.0	31.0	30.0	30.0	31.0	30.4	30.0
90th percentile	31.0	31.0	31.0	31.2	31.0	31.0	31.2	31.0	30.3	31.0	31.0	31.2
Maximum	38.0	31.0	33.0	33.0	32.0	32.0	32.0	32.0	32.0	32.0	33.0	31.2
Mean	29.7	29.8	29.5	29.8	30.1	29.6	30.3	30.2	29.7	30.4	28.8	30.1

Physico-chemical water quality parameters

Concentration levels of the inorganic ions and organic carbon contents differed between the individual components, with highest concentrations measured for calcium and lowest for nitrite (Table S5). Across the study period, TOC, DOC, chloride, sulfate and calcium concentrations showed tendencies of seasonal patterns, with slight maxima in summer and early autumn 2018 and lower concentrations during winter 2018/2019 (Figure S6). However, concentrations of fluoride, nitrate and potassium tend to decrease over the study period, whereas phosphate concentrations showed general increasing trends from November 2018 onwards. The concentrations of sodium and magnesium were rather constant during the study period; concentrations of ammonium and nitrite were mostly below the limit of quantification (LOQ), with, however, distinct increased levels particularly during summer and autumn 2018 (Figure S6). Importantly, the concentrations of most components showed a rather high inter-pond variability at the individual sampling dates. However, no pond showed generally higher or lower concentrations across the study period when considering all water chemistry components (Table S5).

The specific conductivity averaged across all ponds was generally between 100 and 200 $\mu\text{S}/\text{cm}$ during the study period (Figure S7A). The conductivity increased from low levels in winter 2017/2018 (mean conductivity across all ponds of 97.8 $\mu\text{S}/\text{cm}$; 95% CI [95.5, 100.2]) to a maximum value at the beginning of May 2018 (mean of 266 $\mu\text{S}/\text{cm}$; 95% CI [255.1, 277.1]) and subsequently decreased to lower values in autumn and winter 2018. Subsequent to lower specific conductivities in May 2019 (mean of 101 $\mu\text{S}/\text{cm}$; 95% CI [83.9, 117.9]), an increase till the end of July 2019 was observed (Figure S7A). In November 2017 practically no variation existed between the ponds. However, the variation between the FPMs increased over time with differences of $>200 \mu\text{S}/\text{cm}$ at the end of the study period (Figure S7A). Based on mean and median values (Table S6), FPM8, FPM9, FPM10, FPM12 had lower specific conductivities (median $\leq 120 \mu\text{S}/\text{cm}$), whereas FPM1, FPM3, FPM4, and FPM7 were characterized by higher specific conductivities (mean and median $> 150 \mu\text{S}/\text{cm}$). However, apart from FPM4 with a maximum of 330 $\mu\text{S}/\text{cm}$, specific conductivity did not exceed 300 $\mu\text{S}/\text{cm}$ in the other ponds (Figure S7A; Table S6).

The pH values averaged across the twelve FPMs were mostly in the range of 7.5 to 9.5 during the entire study period (Figure S7B). Despite some variations, the pH values showed an generally increasing trend from November 2017 (mean pH across all ponds of 7.9; 95% CI [7.7, 8.1]) to a maximum in September 2018 (mean pH of 9.6; 95% CI [9.4, 9.9]). In 2019, average pH values were rather constant between 8.5 and 9 but decreased in July 2019 to values between 7.5 and 8 (Figure S7B). The individual pH values of the FPMs were generally in the range between 7.2 and 10 during the study period (Figure S7B). However, a rather large variation (i.e., up to more than 2.5 pH unit difference) was recorded between individual FPMs for individual sampling dates, particularly from April 2019 onwards. Based on mean and median pH values of the individual ponds, slightly higher pH values were recorded in FPM2, FPM5 and FPM 8-12 (median and mean pH ≥ 8.5) compared with the other ponds (Table S6).

The water temperature in 25 cm depth showed a clear seasonal pattern, with lowest temperatures in winter months and highest temperatures during summer (Figure S7C). Temperatures did not fall below 2°C and the maximum temperature did not exceed 26.1°C (Table S6). The water temperature did not differ largely between the individual FPMs (Figure S7C). However, care has to be taken when comparing the temperatures of the FPM9-FPM12 due to missing values for individual months (see Table S6).

The dissolved oxygen concentrations increased from the beginning of November till end of December 2017 and dropped from March 2018 to low oxygen concentrations in spring and summer 2018 (Figure S7D); a subsequent increase of dissolved oxygen concentrations till February 2019 was followed by a decrease till the end of July 2019. The dissolved oxygen concentrations showed thus an opposing trend to water temperature, with higher dissolved oxygen concentrations in winter months and lower during summer months. The variation of dissolved oxygen concentrations between the individual FPMs increased over time with particularly high variations (i.e., differences of up to $>25 \text{ mg}/\text{L}$) occurring in winter 2018/2019 (Figure S7D). Excepting FPM8, which had generally higher dissolved oxygen concentrations, no large differences exist between the median and mean oxygen concentrations of the FPMs

(Table S6). Maxima of dissolved oxygen concentrations were generally <20 mg/L, with, however, maximum concentrations of >20 mg/l (up to 26.8 mg/L) were recorded for FPM2, FPM8, and FPM 10-12 (Figure S7D; Table S6). Again, care have to be taken in interpreting the oxygen concentrations of individual FPMs due to logger failures particularly for FPM 9-12 (Table S6).

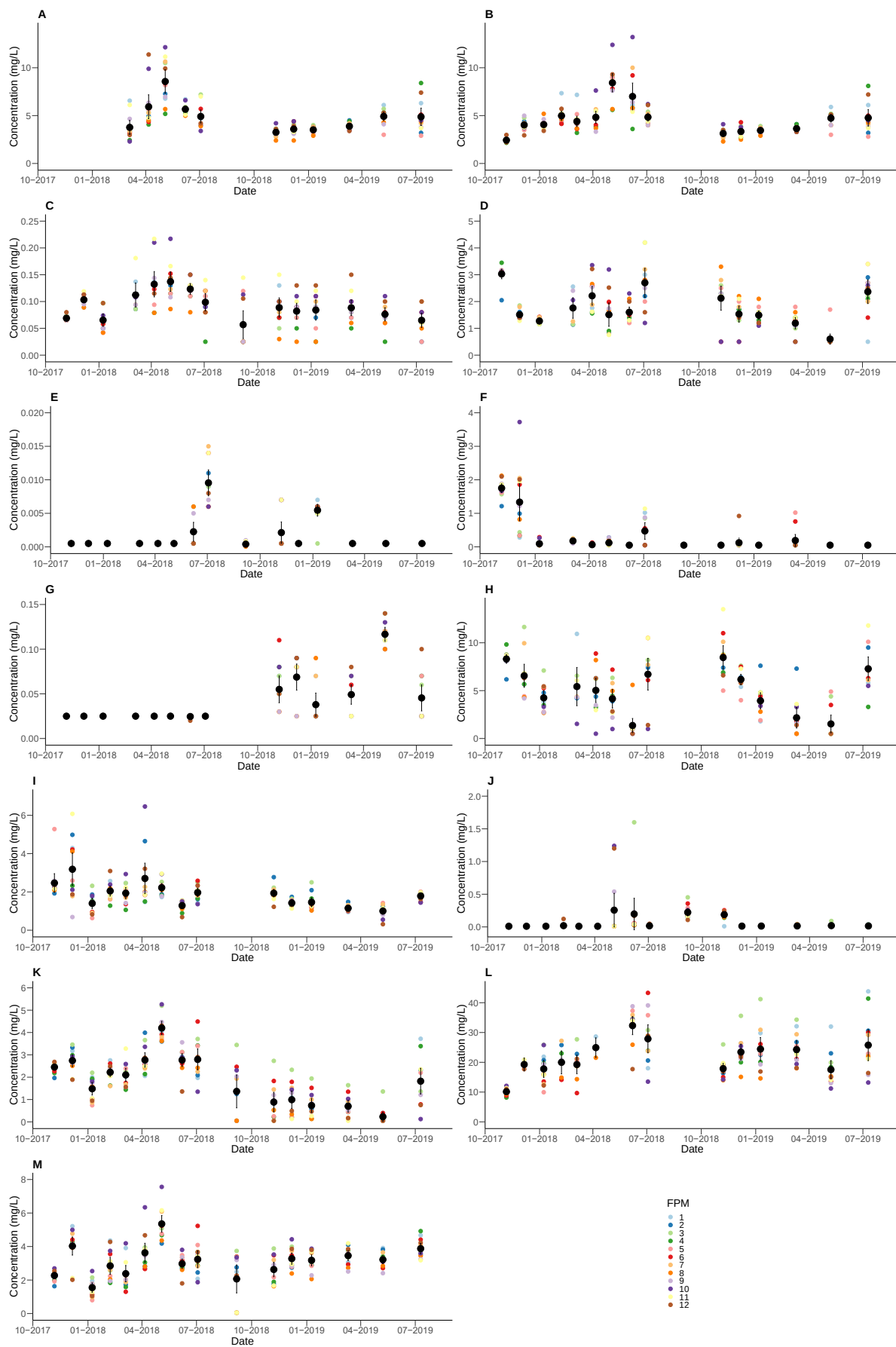


Figure S6. Concentrations of (A) TOC, (B) DOC, (C) fluoride, (D) chloride, (E) nitrite, (F) nitrate, (G) phosphate, (H) sulfate, (I) sodium, (J) ammonium, (K) potassium, (L) calcium, and (M) magnesium in the twelve FPMs over time (6 November 2017 – 31 July 2019). Black circles depict means and error bars show 95% confidence intervals.

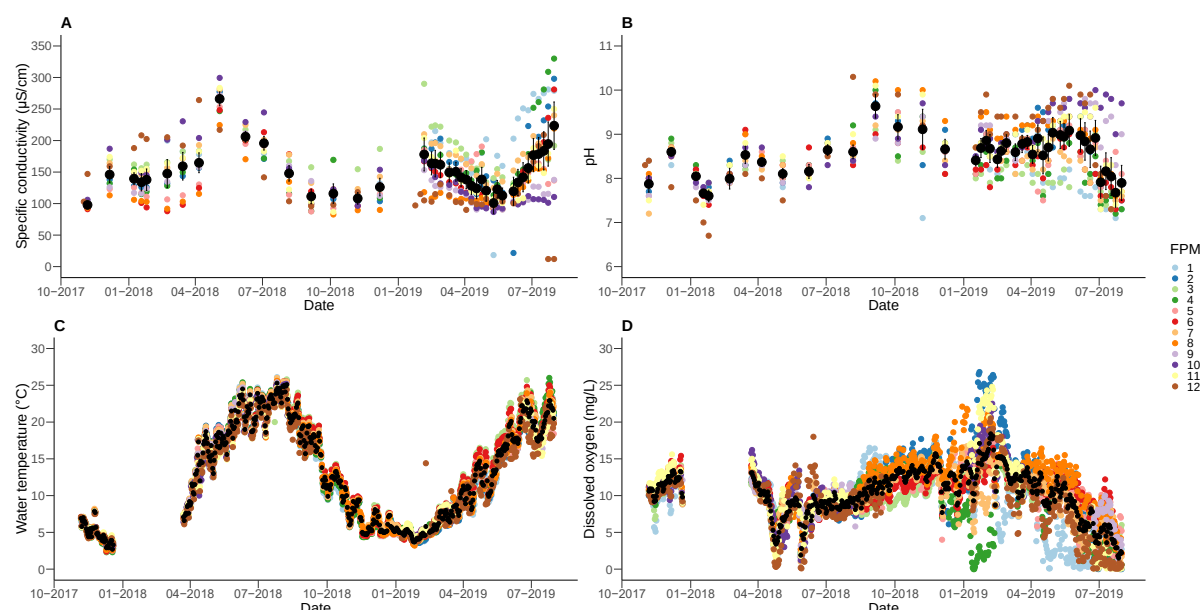


Figure S7. Development of (A) specific conductivity, (B) pH-value, (C) water temperature, and (D) dissolved oxygen concentration for FPM 1-12 over time (6 November 2017 – 31 July 2019). Black circles depict means and error bars show 95% confidence intervals. Missing values are due to equipment failure (see Table S6).

Table S5. Statistical characteristics of the water chemistry (i.e., concentrations of anions, cations, total organic carbon (TOC), dissolved organic carbon (DOC)) for the study period (06 November 2017 – 31 July 2019). Concentrations less than the limit of quantification (LOQ) were replaced by 0.5 x LOQ.

TOC (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	3.0000	3.1000	3.4201	2.4263	2.9000	3.1832	2.8000	2.4000	3.1000	2.2962	3.3000	2.9805
25th percentile	4.0500	3.4244	4.0000	3.6500	3.1500	3.8500	3.6000	3.2998	3.4000	3.4500	3.5500	3.5500
Median	4.7000	4.0000	4.6000	4.2363	3.4000	4.4000	4.2000	3.9000	4.1000	4.4000	4.7837	4.2000
Mean	5.1253	4.3792	5.3048	4.5581	4.1331	4.7738	4.8498	3.9166	4.5060	5.4022	5.1307	5.5796
75th percentile	6.4308	5.1252	5.8000	5.1956	4.5500	5.3500	5.1401	4.4495	5.3500	5.8500	5.6029	6.6000
90th percentile	6.7000	5.5000	7.2000	5.6100	5.6426	5.7000	5.8000	5.0000	6.3440	9.8919	7.0000	9.9154
Maximum	6.7837	7.2720	10.486	8.4000	8.1830	8.3899	10.638	5.6845	6.9584	12.136	11.148	11.380
DOC (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	2.6106	2.3183	2.4602	2.1623	2.3270	2.2302	2.4564	2.2737	2.3888	2.4585	2.5409	2.9405
25th percentile	3.9000	3.5000	3.8827	3.2996	3.1000	3.6430	3.7500	3.2779	3.3000	3.7559	3.4500	3.4038
Median	4.7000	4.1622	4.4000	3.9309	3.4914	4.1783	4.1672	3.8920	4.0000	4.1590	3.7670	4.1898
Mean	5.1137	4.3169	4.6641	4.1049	4.0766	4.6011	4.7769	3.8994	4.2071	5.5283	4.3278	4.8301
75th percentile	6.1000	4.5318	5.0434	4.3832	4.5267	4.9000	5.0500	4.4695	4.7776	5.6853	4.9815	5.9041
90th percentile	7.2727	5.5200	5.6400	5.4023	5.4218	6.7328	7.4327	5.4811	5.8366	10.484	5.4481	7.1600
Maximum	8.9608	8.4478	9.1169	8.1000	8.9780	9.2000	10.000	5.8000	7.6200	13.200	8.5700	9.3100
Fluoride (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0728	0.0600	0.0800
25th percentile	0.0800	0.0700	0.0650	0.0313	0.0613	0.0725	0.0800	0.0293	0.0675	0.0950	0.1000	0.1000
Median	0.0900	0.0863	0.0800	0.0625	0.0818	0.0950	0.0935	0.0600	0.0900	0.1100	0.1200	0.1091
Mean	0.0959	0.0820	0.0862	0.0641	0.0808	0.0928	0.0933	0.0566	0.0854	0.1171	0.1278	0.1140
75th percentile	0.1300	0.1004	0.1134	0.0928	0.1077	0.1175	0.1134	0.0798	0.0975	0.1129	0.1473	0.1300
90th percentile	0.1371	0.1259	0.1299	0.1073	0.1137	0.1218	0.1270	0.0844	0.1152	0.1859	0.1749	0.1483
Maximum	0.1444	0.1299	0.1444	0.1153	0.1195	0.1517	0.1300	0.0890	0.1444	0.2170	0.2170	0.1500
Chloride (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.5000	0.5000	0.5000	0.5000	1.2000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

25th percentile	1.2945	1.1793	1.1890	1.2644	1.7000	1.2000	1.2979	1.6000	1.4013	0.6500	1.2867	1.2593
Median	1.6000	1.5986	1.3023	1.4143	1.9965	1.4289	1.6995	2.1000	1.6699	1.3112	1.6081	1.6000
Mean	1.6822	1.5738	1.7396	1.5994	2.0354	1.5540	1.7731	2.1131	1.8665	1.6818	1.8816	1.7908
75th percentile	2.0897	1.9617	2.1127	1.8000	2.1962	1.9775	1.9706	2.7000	2.4763	2.5250	2.1750	2.5000
90th percentile	2.8671	2.2000	2.9716	2.2800	3.0338	2.7553	2.9295	3.0549	2.6515	3.1717	3.3047	2.9794
Maximum	3.0093	2.9000	4.2000	3.4469	3.4000	3.0214	3.2000	3.3000	3.1430	3.3561	4.2000	3.2135
Nitrite (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0005	0.0005	0.0002	0.0001	0.0001	0.0001	0.0004	0.0001	0.0005	0.0004	0.0005	0.0005
25th percentile	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Median	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Mean	0.0019	0.0016	0.0014	0.0008	0.0012	0.0022	0.0018	0.0016	0.0016	0.0017	0.0022	0.0014
75th percentile	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0007	0.0005	0.0007	0.0005
90th percentile	0.0066	0.0038	0.0038	0.0005	0.0038	0.0066	0.0038	0.0060	0.0056	0.0060	0.0062	0.0038
Maximum	0.0090	0.0110	0.0090	0.0050	0.0060	0.0140	0.0150	0.0060	0.0070	0.0070	0.0140	0.0080
Nitrate (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500
25th percentile	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500
Median	0.0500	0.0500	0.0500	0.0500	0.0500	0.0883	0.0500	0.0500	0.0500	0.0500	0.0500	0.0500
Mean	0.2509	0.2468	0.2372	0.2097	0.2519	0.4107	0.3094	0.2636	0.3280	0.4341	0.3786	0.4076
75th percentile	0.1007	0.2386	0.1007	0.0500	0.0500	0.4861	0.0500	0.0500	0.1944	0.1515	0.1132	0.1448
90th percentile	0.7237	0.8025	0.6839	0.2899	0.8151	1.4113	1.1441	0.6331	1.0835	1.1060	1.5713	1.6846
Maximum	1.7633	1.2143	1.5724	1.7872	1.6201	1.8578	2.0503	2.1214	1.8827	3.7252	1.9541	2.0975
Phosphate (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0200
25th percentile	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250
Median	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250	0.0250
Mean	0.0486	0.0354	0.0468	0.0471	0.0404	0.0458	0.0377	0.0358	0.0311	0.0443	0.0368	0.0504
75th percentile	0.0700	0.0250	0.0675	0.0725	0.0500	0.0600	0.0250	0.0250	0.0250	0.0588	0.0250	0.0800
90th percentile	0.0840	0.0640	0.0840	0.0800	0.0780	0.1040	0.0660	0.0770	0.0250	0.0870	0.0710	0.0980
Maximum	0.1200	0.1000	0.1400	0.1100	0.1000	0.1200	0.1200	0.1000	0.1100	0.1300	0.1100	0.1400
Sulfate (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.5000	0.5000	1.1000	0.5000	0.5000	1.2000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
25th percentile	2.3106	4.2473	5.0077	2.8609	2.7805	4.4000	3.6444	2.8000	2.3250	1.1250	3.1273	1.4000
Median	5.5311	6.2883	6.8301	3.9655	4.4105	6.3000	5.5880	5.6000	4.0457	3.3058	5.4298	5.4501
Mean	4.8858	5.4993	6.5686	4.2477	4.5303	6.0042	5.2774	5.4106	4.3215	3.6518	6.0352	4.3439
75th percentile	6.6774	7.3750	8.0500	5.7545	5.0000	7.5000	7.3000	8.1910	6.5250	6.0250	8.3354	6.3089
90th percentile	7.9312	7.5400	8.7953	6.8100	7.9763	8.7295	9.4531	8.5983	7.8319	7.7308	11.410	7.0000
Maximum	10.927	9.5000	11.649	9.8214	10.100	11.000	10.100	10.500	8.5000	8.6954	13.500	8.3842
Sodium (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.9060	1.2300	1.1200	0.8890	0.6306	0.9495	0.9039	1.0300	0.6864	0.5520	1.1300	0.3120
25th percentile	1.2750	1.6800	1.7800	1.2549	1.3550	1.2870	1.4550	1.2404	1.3750	1.4000	1.2450	1.1200
Median	1.4900	1.8780	2.2204	1.5303	1.8900	1.4600	1.6960	1.8459	1.5400	1.7922	1.8036	1.6300
Mean	1.7687	2.2412	2.1323	1.5455	1.9731	1.7864	1.7041	1.8027	1.6006	2.1117	2.0077	1.6876
75th percentile	1.9677	2.1062	2.4950	1.8322	2.2207	2.0710	2.0064	2.0493	1.9350	2.3184	2.1284	2.3075
90th percentile	2.5225	3.8988	2.5719	2.0570	2.6053	2.4277	2.1674	2.1531	2.1818	2.9274	2.6618	2.8023
Maximum	4.2879	4.9805	3.1546	2.3234	5.2796	4.2124	2.2635	4.1116	2.5733	6.4609	6.0761	3.2058
Ammonium (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100
25th percentile	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100
Median	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100
Mean	0.0252	0.0306	0.1544	0.0319	0.0473	0.0512	0.0434	0.0303	0.0652	0.1309	0.0347	0.1243
75th percentile	0.0178	0.0100	0.0503	0.0175	0.0138	0.0235	0.0185	0.0100	0.0185	0.0765	0.0100	0.1107
90th percentile	0.0495	0.0899	0.3111	0.0972	0.1600	0.1505	0.1430	0.0885	0.1675	0.2383	0.0974	0.2050
Maximum	0.1435	0.1800	1.6000	0.1702	0.2912	0.3579	0.2448	0.1782	0.5445	1.2382	0.2300	1.2027
Potassium (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.0500	0.0500	1.3600	0.0500	0.0500	0.3940	0.2090	0.0500	0.0500	0.0500	0.0500	0.0500
25th percentile	0.7880	0.9863	2.1344	0.2163	0.2450	1.4775	1.3375	0.2975	1.2075	1.1575	0.1965	0.4790
Median	2.0253	2.0012	2.5042	1.7782	2.1482	2.0473	1.7658	1.9531	1.8902	2.0246	2.0554	1.3600
Mean	1.9666	1.9292	2.7135	1.6986	1.6663	2.1317	1.8339	1.5500	1.9634	2.0022	1.7376	1.5319
75th percentile	2.8378	2.4650	3.4464	2.7484	2.6247	2.6100	2.3956	2.4260	2.6423	2.6627	2.6819	2.5851
90th percentile	3.4486	3.4627	3.6835	3.4050	3.2600	3.3198	2.7337	2.4913	3.1500	3.1400	3.0930	2.8867
Maximum	4.0520	3.9928	5.1978	4.2891	3.6965	4.4900	3.7952	3.6174	4.4669	5.2570	4.2101	4.3088
Calcium (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12

Minimum	10.596	9.7351	10.968	8.1792	8.9930	8.8255	9.7111	9.3401	10.740	11.200	11.327	11.004
25th percentile	19.935	20.687	24.280	19.131	16.495	13.600	18.391	14.600	16.775	13.275	17.725	16.050
Median	28.639	22.651	27.714	20.050	20.912	21.100	25.100	15.300	19.542	16.800	21.200	17.300
Mean	26.628	22.770	28.167	22.346	21.304	21.171	23.357	17.665	21.307	19.348	21.447	17.816
75th percentile	31.900	26.038	34.550	23.375	23.525	26.100	29.250	21.528	20.900	25.125	24.100	18.864
90th percentile	32.100	30.220	35.600	31.370	34.760	32.320	30.750	25.120	37.040	26.646	28.100	21.736
Maximum	43.800	32.200	41.200	41.400	37.300	43.300	35.900	28.800	39.100	34.400	34.000	28.900
Magnesium (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	1.9554	1.5928	2.1506	0.0500	0.0500	1.1256	0.9722	0.0500	1.7741	1.8700	0.0500	1.0419
25th percentile	2.6181	1.8724	2.8473	1.8750	2.2235	2.6965	2.7752	2.0320	2.3243	3.2901	2.4048	2.2779
Median	2.9850	2.7493	3.6900	2.8800	2.9600	3.2898	3.4948	2.5000	2.8800	3.6703	3.0861	3.4600
Mean	3.3570	2.8138	3.4236	2.7781	2.8277	3.2772	3.3852	2.5111	2.9590	3.9463	2.9734	3.2231
75th percentile	4.0224	3.6025	3.8966	3.4950	3.5075	3.7655	3.7927	2.9100	3.3781	4.2531	3.4975	3.9500
90th percentile	4.9414	4.0700	4.1624	4.4616	4.1195	4.8200	4.4403	3.6905	3.8249	5.6670	4.0106	4.4749
Maximum	5.3401	4.1826	5.0813	4.9200	4.7645	5.2377	6.0803	4.3554	5.4063	7.5609	6.1586	5.3762

The limits of quantification (LOQ) were as follows: TOC: 0.5 mg/L; DOC: 0.5 mg/L; Fluoride: 0.05 mg/L; Chloride: 1 mg/L; Nitrite: 0.001 mg/L; Nitrate: 0.1 mg/L; Phosphate: 0.05 mg/L; Sulfate: 1 mg/L; Sodium: 0.1 mg/L; Ammonium: 0.02 mg/L; Potassium: 0.1 mg/L; Calcium: 0.1 mg/L; Magnesium: 0.1 mg/L.

Table S6. Statistical characteristics of the physico-chemical parameters specific conductivity, pH, water temperature and dissolved oxygen measured for FPM 1-12 during the study period (06 November 2017 – 31 July 2019).

Specific conductivity (µS/cm)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	18.3	21.6	98.3	88.1	87.5	87.9	98.6	82.7	90.4	86.6	86.6	90.0
25th percentile	132.0	122.5	162.0	115.5	110.1	100.3	145.2	97.2	120.5	139.5	122.7	103.5
Median	179.9	143.5	166.6	151.1	131.9	130.0	169.8	108.4	120.3	117.5	146.9	115.0
75th percentile	206.0	176.5	188.4	177.4	160.5	159.1	193.0	128.2	135.0	154.2	174.0	171.9
90th percentile	274.1	223.9	222.0	252.9	185.6	201.4	207.3	170.5	158.0	206.5	189.6	205.5
Maximum	281.0	298.0	290.0	330.0	253.4	281.0	281.4	224.5	272.1	299.4	283.5	264.2
Mean	182.8	155.8	174.5	163.1	140.5	142.3	170.3	121.3	126.5	137.0	149.6	137.8
pH	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	7.1	7.3	7.4	7.2	7.5	7.3	7.2	7.6	7.5	7.8	7.3	6.7
25th percentile	8.1	8.3	8.1	8.0	8.1	8.1	8.2	8.2	8.0	8.1	8.4	8.0
Median	8.2	8.7	8.1	8.3	8.6	8.3	8.4	9.0	8.8	8.9	8.8	9.0
75th percentile	8.4	9.1	8.3	8.8	8.8	8.7	8.6	9.3	9.3	9.5	9.2	9.6
90th percentile	8.9	9.4	8.6	9.0	9.1	8.9	8.9	9.4	9.7	9.7	9.4	9.9
Maximum	9.7	9.7	9.1	10.0	9.9	9.1	9.3	10.2	9.8	10.0	10.1	10.3
Mean	8.1	8.6	8.1	8.4	8.5	8.3	8.3	8.8	8.8	8.9	8.7	8.8
Temperature (°C)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	2.6	2.6	2.5	2.3	2.4	2.4	2.7	2.6	6.7	2.9	3.4	3.3
25th percentile	6.4	6.5	6.6	6.5	6.5	6.6	6.8	6.5	16.8	5.4	5.9	7.3
Median	11.7	12.4	13.6	12.8	13.1	13.4	12.9	12.8	19.7	11.3	15.5	15.3
75th percentile	19.1	19.1	19.5	19.4	18.6	19.4	19.2	18.7	22.3	19.7	20.2	18.9
90th percentile	22.6	22.6	22.9	23.0	21.9	22.8	22.6	22.2	23.6	22.7	22.5	21.9
Maximum	26.1	25.6	25.8	26.0	25.5	25.7	26.0	25.6	25.4	25.2	24.7	24.5
Mean	12.9	13.1	13.6	13.3	13.1	13.5	13.3	13.1	18.9*	12.6*	13.6*	13.7*
Oxygen (mg/L)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
Minimum	0.1	0.1	2.0	0.1	1.4	1.5	0.6	2.2	1.2	1.2	0.1	0.1
25th percentile	6.5	8.2	8.3	7.4	8.7	8.9	8.3	9.0	6.9	8.6	7.8	6.7
Median	9.9	10.9	9.8	10.1	10.7	10.9	10.5	12.2	8.6	10.6	9.9	9.3
75th percentile	12.4	14.0	11.4	12.6	12.5	12.1	12.6	14.1	10.0	13.1	13.2	11.4
90th percentile	14.6	16.3	12.9	13.8	14.4	13.4	14.2	15.8	10.6	16.8	17.2	13.2
Maximum	17.5	26.8	19.5	15.9	18.5	16.5	19.0	22.1	12.1	22.5	24.8	20.5
Mean	9.3	11.1	9.9	9.4	10.6	10.5	10.2	11.8	8.2*	11.0*	10.2*	8.7*

Please note that water temperature and dissolved oxygen data are missing due to logger failure (defective devices) for the following FPMs and time spans: FPM 1-12: 22 December 2017 – 22 March 2018; FPM9: 06 November 2017 – 20 December 2017 & 10 August 2018 – 20 June 2019; FPM10: 10 August 2018 – 07 January 2019 & 22 March 2019 – 31 July 2019; FPM11: 10 August 2018 – 07 January 2019 & 22 March 2019 – 20 June 2019; FPM12: 15 September 2018 – 10 January 2019. The mean values for water temperature and dissolved oxygen of FPM 9-12 (indicated by an asterisk) should be interpreted with care due to high amounts of missing values.

Colonization of aquatic and terrestrial ecosystem compartments

Merolimnic insect emergence

Overall, 3345 emerged individuals have been captured in 2019 and identified at the family level; the individuals belong to nine families (*Chironomidae*, *Baetidae*, *Coenagrionidae*, *Culicidae*, *Chaoboridae*, *Libellulidae*, *Tipulidae*, *Sialidae*, *Limnephilidae*) from six orders. The numbers of emerged families increased across all ponds from end of April 2019 (mean of 0.7 families) till mid of June 2019 (mean number of 2.1 families), followed by a decrease till end of July 2019 (mean number of 0.9 families) (Figure S8). Concerning individual FPMs, the highest richness of emerged families ($n = 5$) per sampling date was recorded for FPM1 at the end of June 2019, while, in contrast, a maximum of one emerged family only could concurrently be found for eight of the FPMs.

Across all sampling dates, FPM7 had the overall highest family richness ($n = 8$), whereas only two different families were recorded for FPM8 and FPM12 (Table S7). The total number of individuals also differed substantially between the ponds, with highest numbers found for FPM3 ($n = 843$ individuals) and lowest numbers found for FPM12 ($n = 134$ individuals) (Table S7). Individuals belonging to the families *Chironomidae* (81.2%), *Baetidae* (16.04%) and *Coenagrionidae* (2.14%) were most often detected.

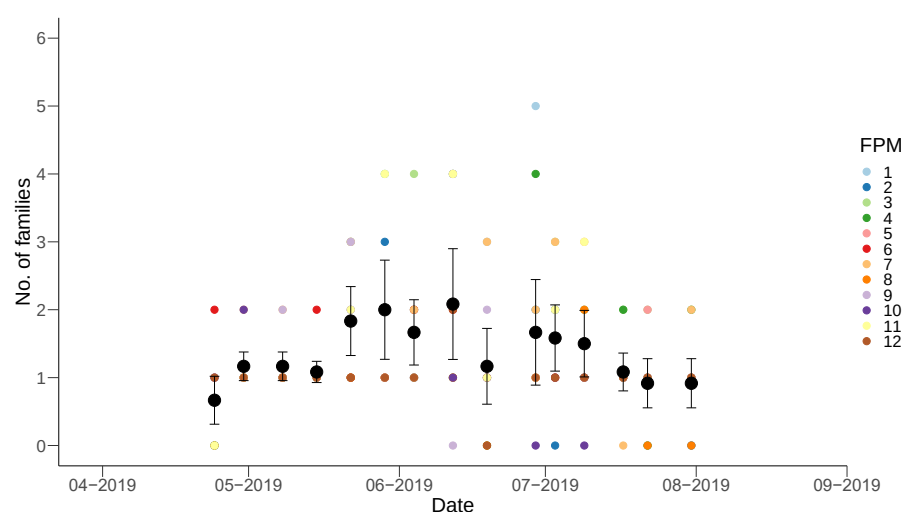


Figure S8. Development of the no. of families of emerged merolimnic insects sampled in FPM 1-12 in 2019. Black circles depict means and error bars show 95% confidence intervals.

Table S7. Detailed results of the macroinvertebrate, merolimnic insect emergence, zooplankton, amphibian and submerged vegetation assessments for FPM 1-12 during the study period (06 November 2017 – 31 July 2019).

No. of macroinvertebrate families	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
11 December 17	2	2	0	0	0	1	0	1	1	0	1	0
05 February 18	2	2	1	1	1	0	2	1	1	2	1	3
02 May 18	3	2	4	3	3	3	3	2	3	2	2	2
29 May 18	1	3	4	3	2	3	3	2	4	1	3	2
27 June 18	7	2	2	5	5	4	4	2	2	4	2	6

01 August 18	7	5	6	4	7	7	4	4	5	7	1	11
22 August 18	9	6	6	5	4	7	6	6	6	8	8	10
24 September 18	7	4	10	4	3	7	8	4	8	6	3	7
18 October 18	4	3	6	4	3	7	4	4	1	3	3	4
16 November 18	6	3	8	2	4	8	3	6	4	7	4	7
14 December 18	5	2	8	5	3	10	5	5	8	5	3	5
11 January 19	5	6	6	4	2	4	8	4	4	11	2	3
15 March 19	4	4	6	5	4	7	6	5	3	5	4	7
02 May 19	5	2	8	4	4	10	4	5	6	7	3	4
27 June 19	6	5	7	3	4	8	6	5	3	4	4	7
Total no. of different families (2017 - 2019)	17	14	17	14	13	17	17	13	17	17	11	19
Total no. of individuals (2017 - 2019)	841	421	1052	417	332	693	451	446	614	806	431	1494
No of emerged families	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
24 April 19	0	0	1	0	1	2	1	0	1	1	0	1
30 April 19	1	1	1	1	1	1	2	1	1	2	1	1
08 May 19	1	1	1	1	1	1	2	1	2	1	1	1
15 May 19	1	1	1	1	1	2	1	1	1	1	1	1
22 May 19	3	1	3	1	1	1	3	1	3	2	2	1
29 May 19	4	3	4	1	1	1	1	1	1	2	4	1
04 June 19	2	1	4	2	2	2	2	1	1	1	1	1
12 June 19	4	4	1	1	1	2	4	1	0	1	4	2
19 June 19	3	1	1	0	1	1	3	0	2	1	1	0
29 June 19	5	2	1	4	1	1	2	1	1	0	1	1
03 July 19	2	0	3	2	1	1	3	1	2	1	2	1
09 July 19	2	1	2	1	1	1	3	2	1	0	3	1
17 July 19	1	1	2	2	1	1	0	1	1	1	1	1
22 July 19	0	1	2	0	2	1	1	0	1	1	1	1
31 July 19	0	0	1	2	1	1	2	0	1	1	1	1
Total no. of different families in 2019	5	4	5	4	3	3	8	2	3	3	5	2
Total no. of individuals in 2019	437	159	843	152	272	167	436	136	221	190	196	134
No. of zooplankton individuals per sample	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
15 November 17	0			13	0		11	17	1	1		2
18 December 17				0	5	3	1	0	0	20	0	1
12 March 18		1		4	3	0	4	18	0	48	1	1
05 April 18	10	8		2	2	3	17	12	71		75	3
28 May 18			0	0	0	0	0	0				1
26 June 18	13	482	1021	162	9	8	8		186	393	972	4048
11 July 18	5	525	3	0	19	8	0	8	76	6896	127	6800
21 August 18	656	840	72	2160	2224	163	1120	10560	1408	1600	792	628
17 September 18	576	2260	31	115	493	25	520	1336	264	572	5056	1808
02 October 18	26	1184	54	50	1752	92	868	1624	588	416	564	296
06 November 18	32	45	68	624	112	100	390	30	252	134	596	72
04 December 18	41	650	14	137	900	5	140	2996	564	68	672	65
17 January 19		38	5	142	304	28	100	224		58	52	48
11 February 19	4	34	36	20	25	46	92	128	352	256	816	384
12 March 19	456	59	28	30	376	81	27	21	124	384	688	1632
29 April 19	2272	968	42	864	21	176	1312	1376	992	1608	1120	1856
23 May 19	3040	3392	160	2176	2688	70	832	2816	784	2368	0	2112
11 June 19	640	2560	1280	496	384	1920	5376	1792	6016	3520	4224	3072
10 July 19	832	3456	640	2496	704	320	992	1184	1120	1344	2048	2816
Overall mean	574	1031	230	500	527	169	622	1341	753	1158	1047	1350
Tadpole coverage (%)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
19 April 18	7.66	29.52	10.61	6.66	5.25	9.21	20.25	6.93	9.85	11.40	26.16	23.02
03 May 18	41.50	19.55	28.01	18.84	23.83	37.29	33.73	31.60	41.14	14.44	20.10	23.60
22 May 18	20.92	3.80	19.51	17.72	29.56	30.60	33.92	25.24	28.65	6.44	0.00	1.10
07 June 18	8.15	0.00	14.34	9.24	19.71	11.16	32.27	45.48	29.17	38.17	3.15	1.45

22 March 19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
26 March 19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
03 April 19	2.92	9.28	2.33	15.74	6.37	2.83	0.00	0.64	0.68	0.00	1.32	3.64
05 April 19	2.58	9.48	0.00	2.50	20.49	2.18	2.75	7.21	0.86	0.00	4.00	3.77
12 April 19	9.57	14.70	0.00	47.25	27.61	1.02	0.00	0.00	0.00	13.63	6.65	5.96
15 April 19	26.70	25.45	8.06	26.77	70.13	2.90	5.18	8.50	3.73	9.21	10.66	2.26
18 April 19	24.36	2.21	4.03	43.36	68.54	0.00	1.51	7.15	0.62	9.57	7.31	13.89
23 April 19	76.53	16.96	18.96	34.18	71.24	0.00	3.80	14.35	4.34	3.69	8.91	5.05
26 April 19	90.59	49.74	17.47	40.16	79.04	0.00	2.20	54.42	5.16	9.53	14.56	17.06
29 April 19	87.45	37.45	34.20	25.77	59.44	0.00	17.14	79.56	42.00	12.32	38.96	27.58
03 May 19	84.44	59.05	65.24	36.33	73.96	0.00	7.45	54.70	44.32	9.39	42.42	30.72
07 May 19	67.66	49.11	31.83	42.87	45.64	0.33	0.00	40.52	29.01	10.61	31.16	28.02
10 May 19	69.03	34.66	53.82	42.80	54.42	0.00	11.63	53.13	36.97	14.69	38.43	35.61
17 May 19	39.55	39.35	37.23	37.39	39.49	0.00	0.00	42.24	32.80	14.89	35.40	34.09
23 May 19	43.76	34.03	36.79	33.49	32.72	0.00	34.91	38.97	25.76	12.89	41.98	44.01
31 May 19	8.11	0.00	36.15	12.10	18.20	0.00	0.00	10.36	27.67	15.30	0.00	0.00
07 June 19	33.29	22.32	25.74	29.31	25.68	27.82	28.88	32.94	0.00	6.64	0.00	12.98
14 June 19	42.80	0.07	0.08	0.24	0.85	0.00	0.47	0.47	0.31	2.09	0.00	0.08
Overall mean	35.80	20.76	20.20	23.76	35.10	5.70	10.73	25.20	16.50	9.77	15.05	14.27
Submerged vegetation coverage (%)	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
31 January 18	2.10	7.85	5.49	1.09	8.61	5.84	8.58	4.56	4.37	12.59	10.63	6.02
20 February 18	0.00	10.69	1.36	1.26	9.35	5.25	11.93	9.73	3.37	7.83	17.08	10.14
08 March 18	0.00	4.33	0.74	1.46	7.21	2.72	6.79	4.70	3.29	27.83	13.90	4.67
22 March 18	3.30	0.00	0.95	1.73	5.32	0.00	3.15	2.23	0.10	6.87	17.54	0.31
05 April 18	0.44	0.78	1.04	0.25	1.50	0.00	3.25	1.15	0.00	1.63	2.37	0.00
03 May 18	0.00	0.00	0.00	8.42	0.00	0.00	0.00	0.00	0.00	11.37	0.00	0.00
22 May 18	0.00	1.56	0.00	0.67	15.26	0.00	0.00	0.00	4.78	1.69	0.00	11.43
08 June 18	4.35	7.31	1.60	4.26	10.65	2.90	5.61	2.84	3.47	1.29	0.10	0.00
13 July 18	19.33	30.65	3.55	27.74	20.87	22.83	12.28	17.34	21.30	6.55	14.23	13.02
20 August 18	44.67	36.09	9.17	57.86	44.78	19.36	30.83	53.80	35.02	5.33	6.46	45.08
27 August 18	44.82	36.80	8.13	60.12	49.91	16.85	28.41	57.34	34.65	22.24	59.43	50.37
31 August 18	48.40	50.95	11.29	75.69	56.38	23.73	31.65	69.76	43.12	26.11	63.05	59.95
10 September 18	48.60	56.31	15.06	79.67	71.93	26.49	36.92	66.60	41.12	22.83	68.67	59.33
28 September 18	58.26	67.75	14.09	75.03	58.50	30.32	35.80	72.36	52.96	34.78	68.48	67.20
29 October 18	58.58	70.70	15.24	72.89	55.88	26.37	40.47	71.65	38.45	36.43	73.04	65.68
16 November 18	55.22	76.52	12.30	67.62	60.95	24.47	36.82	69.18	33.23	32.02	74.59	48.50
30 November 18	52.11	70.00	11.35	73.74	65.21	22.80	34.46	63.16	41.21	35.25	68.50	70.76
21 December 18	57.93	76.97	13.08	73.97	67.49	29.52	33.74	64.36	40.01	40.19	71.60	73.97
04 January 19	61.61	84.65	11.21	74.92	71.27	32.21	41.14	70.16	43.30	42.22	72.89	79.23
08 February 19	46.69	59.01	7.57	64.89	60.67	23.12	34.80	53.82	35.87	37.07	64.62	67.53
20 March 19	63.26	91.81	23.33	82.50	80.31	36.23	40.28	82.20	43.67	43.56	86.74	83.13
03 April 19	51.30	82.39	21.16	69.39	67.55	34.15	45.64	70.77	48.06	45.71	79.40	70.16
17 April 19	58.08	87.93	19.16	69.35	68.32	35.66	50.11	68.47	57.49	46.11	74.76	69.24
30 April 19	60.87	66.49	39.83	73.50	66.19	40.20	49.12	66.55	62.64	44.03	66.85	67.18
15 May 19	68.72	78.56	23.82	67.23	74.16	51.47	54.82	71.64	54.63	49.18	73.78	76.34
29 May 19	84.78	84.20	24.59	70.66	75.96	53.59	56.76	67.48	67.34	52.69	70.00	73.65
12 June 19	81.90	84.72	36.09	80.69	70.93	39.22	57.16	73.67	76.35	59.07	75.09	74.15
25 June 19	86.81	82.54	44.03	76.30	77.51	54.37	69.67	75.53	75.17	69.16	71.43	83.43
10 July 19	84.14	87.63	45.30	86.81	84.14	93.67	80.87	84.30	85.31	80.30	85.62	85.55
23 July 19	89.65	81.04	47.39	87.65	90.49	84.67	88.61	92.92	81.10	83.83	86.70	83.93
Overall mean	44.53	52.54	15.60	52.91	49.91	27.93	34.32	50.28	37.71	32.86	51.25	50.00

Table S8. Results of the terrestrial vegetation survey at the FPMs in November 2018 (species present: 1; species absent: 0).

Species	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
<i>Agrostis stolonifera</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Alisma plantago-aquatica</i>	0	0	0	0	0	0	1	1	0	1	0	1
<i>Alopecurus aequalis</i>	1	1	1	1	1	0	1	0	0	1	1	0
<i>Betula pendula</i>	1	0	0	0	0	0	0	0	0	0	0	0
<i>Carex acutiformes</i>	1	1	0	1	0	0	0	0	0	0	0	0
<i>Carex</i> sp.	0	1	0	0	0	0	0	0	1	0	1	1
<i>Carex</i> sp.	0	1	0	0	1	0	0	0	0	0	0	0
<i>Cerastium fontanum</i>	0	0	0	0	0	0	1	0	0	0	0	1
<i>Chenopodium album</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Cichorium intybus</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Cirsium palustre</i>	0	1	0	0	0	0	0	0	0	1	0	1
<i>Cyperus fuscus</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Digitaria sanguinalis</i>	0	0	0	0	0	0	0	1	0	0	0	0
<i>Echinocloa crus-galli</i>	0	1	0	1	0	0	0	1	1	1	1	1
<i>Epilobium palustre</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Equisetum arvense</i>	1	1	0	0	1	0	1	0	1	0	0	0
<i>Eragrostis minor</i>	0	0	0	0	0	0	0	0	0	0	0	1
<i>Erigeron annuus</i>	0	0	0	1	0	0	1	1	0	0	0	1
<i>Eupatorium cannabinum</i>	0	0	0	0	0	0	0	0	0	0	0	1
<i>Euphorbia cyparissias</i>	0	0	0	0	1	0	0	0	0	0	0	0
<i>Glyceria fluitans</i>	1	1	1	0	0	0	0	1	0	1	0	0
<i>Gnaphalium uliginosum</i>	0	0	0	1	0	0	1	0	1	0	1	0
<i>Holcus lanatus</i>	0	0	0	0	0	0	0	0	0	1	0	0
<i>Isolepis setacea</i>	1	0	1	0	0	1	0	0	1	1	0	1
<i>Juncus effusus</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Juncus tenuis</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Juncus articulatus</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Juncus bufonius</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Lactuca serriola</i>	0	0	0	0	0	0	0	1	0	0	0	1
<i>Lolium perenne</i>	1	1	1	1	0	0	0	1	0	1	0	0
<i>Lycopus europaeus</i>	0	1	0	1	0	0	0	1	0	0	0	0
<i>Lythrum salicaria</i>	0	1	0	0	0	1	1	0	0	0	0	1
<i>Malva sylvestris</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Medicago lupulina</i>	1	1	0	0	1	1	1	1	1	1	1	1
<i>Persicaria maculosa</i>	0	1	0	1	0	0	0	1	1	0	1	1
<i>Plantago lanceolata</i>	0	0	0	1	1	1	1	0	0	1	0	0
<i>Plantago major</i>	0	1	1	1	1	0	1	1	0	0	1	1
<i>Poa annua</i>	0	0	0	0	0	0	0	0	0	0	0	1
<i>Poa compressa</i>	0	0	0	0	0	0	0	0	0	0	0	1
<i>Prunella vulgaris</i>	0	0	0	0	0	0	1	0	0	1	0	0
<i>Ranunculus repens</i>	1	1	1	0	1	0	0	1	0	0	1	0
<i>Salix</i> sp.	1	1	1	1	1	1	1	1	1	1	1	1
<i>Salix caprea</i>	0	1	1	1	0	1	1	0	0	0	0	1
<i>Salix</i> sp.	0	0	1	0	0	0	0	0	0	0	0	0
<i>Solidago canadensis</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Sonchus oleratus</i>	1	1	0	0	0	0	1	0	0	0	0	0
<i>Trifolium pratense</i>	0	1	0	1	0	1	0	0	0	0	0	1
<i>Trifolium repens</i>	1	1	0	0	0	1	0	0	0	0	1	1
<i>Tussilago farfara</i>	0	0	1	0	0	0	0	1	0	0	1	1
<i>Typha latifolia</i>	1	1	1	1	1	1	1	1	1	1	1	1
<i>Urtica dioica</i>	0	0	0	0	1	0	0	0	0	0	0	0
<i>Veronica beccabunga</i>	1	1	1	0	1	1	0	0	0	0	0	0
<i>Vicia hirsuta</i>	1	0	0	0	1	1	0	0	0	0	0	0
Total no. of species	21	33	18	20	19	17	21	21	15	19	18	28

Table S9. Overall results of the ground beetle monitoring at the twelve FPMs for the sampling campaigns in May 2018, September/October 2018 and May 2019.

Species	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
<i>Acupalpis flavicollis</i>	0	0	0	0	0	1	1	1	0	0	0	1
<i>Acupalpus meridianus</i>	0	0	0	0	1	0	0	0	0	0	0	0
<i>Agonum emarginatum</i>	5	0	1	11	1	1	0	6	7	3	2	3
<i>Agonum fuliginosum</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Agonum muelleri</i>	2	0	0	0	0	0	0	0	1	0	0	1
<i>Agonum sexpunctatum</i>	41	10	20	16	16	21	7	26	12	11	14	18
<i>Agonum viridicupreum</i>	0	0	0	1	0	0	0	0	0	0	0	0
<i>Amara aenea</i>	1	0	1	0	0	0	1	0	0	0	0	0
<i>Amara montivaga</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Anisodactylus binotatus</i>	1	0	0	0	0	0	0	0	0	1	0	0
<i>Bembidion articulatum</i>	1	0	0	2	0	1	4	0	1	0	1	1
<i>Bembidion biguttatum</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Bembidion bruxellense</i>	0	0	0	0	0	0	1	0	0	0	0	0
<i>Bembidion femoratum</i>	0	1	1	0	1	1	0	0	0	0	1	0
<i>Bembidion guttula</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Bembidion illigeri</i>	0	1	3	2	9	3	7	9	7	2	9	4
<i>Bembidion lampros</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Bembidion semipunctatum</i>	0	0	1	0	0	0	0	0	0	0	0	0
<i>Bembidion tetracolum</i>	3	3	3	0	1	0	2	0	1	3	0	5
<i>Carabus granulatus</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Chlaenius nigricornis</i>	0	0	0	0	0	0	0	0	1	0	1	1
<i>Chlaenius vestitus</i>	30	23	49	34	32	31	47	62	48	60	46	31
<i>Cicindela campestris</i>	2	0	1	0	1	0	2	1	0	0	0	2
<i>Clivina collaris</i>	0	0	0	1	0	0	0	0	0	0	0	0
<i>Diachromus germanus</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Dyschirius agnatus</i>	0	0	1	1	0	0	1	0	0	0	1	1
<i>Elaphrus cupreus</i>	0	0	0	0	0	0	0	2	2	1	0	3
<i>Elaphrus riparius</i>	2	8	5	2	13	10	7	3	6	6	15	7
<i>Harpalus rufipes</i>	0	0	0	0	0	0	0	1	0	0	0	0
<i>Loricera pilicornis</i>	0	0	0	0	0	0	2	0	0	0	0	0
<i>Nebria brevicollis</i>	9	6	10	3	3	5	11	6	1	2	3	8
<i>Nebria brevicollis</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Oodes helopioides</i>	0	0	0	1	0	0	0	2	0	0	0	1
<i>Panagaeus cruxmajor</i>	0	0	0	0	0	0	0	0	0	0	1	0
<i>Paranchus albipes</i>	1	5	3	4	5	2	4	3	28	20	16	17
<i>Poecilus cupreus</i>	0	1	0	0	0	0	0	0	0	0	0	0
<i>Pterostichus diligens</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Pterostichus minor</i>	0	0	0	0	0	0	0	0	0	0	0	0
<i>Pterostichus nigrata</i>	0	0	0	0	0	0	0	1	1	1	2	4
<i>Stenolophus mixtus</i>	0	0	1	0	0	0	0	0	0	0	0	0
<i>Stenolophus teutonius</i>	35	11	2	11	4	23	24	2	10	13	19	8
<i>Trechus obtusus</i>	0	1	1	0	0	0	0	0	0	0	0	0
Total no.of individuals	133	71	103	89	87	99	121	125	126	123	131	116
Total no. of species	13	12	16	13	12	11	15	14	14	12	14	18

Table S10. Overview of the leafhopper species captured at FPM3, FPM5, FPM8 and FPM10 in June and September 2018.

Species	FPM3	FPM5	FPM8	FPM10
<i>Macrosteles sexnotatus</i>	84	62	44	31
<i>Arthaldeus pascuellus</i>	27	39	18	38
<i>Javesella pellucida</i>	59	15	25	35
<i>Deltocephalus pulicaris</i>	69	12	13	32
<i>Anaceratagallia ribauti</i>	30	17	12	24
<i>Cicadella viridis</i>	2	2	6	12
<i>Muelleriana fairmairei</i>	0	1	1	0
<i>Psammotettix helvolus</i>	18	20	22	2
<i>Psammotettix confinis</i>	9	21	13	12
<i>Zyginidia scutellaris</i>	13	17	3	9
<i>Megophthalmus scanicus</i>	0	0	1	0
<i>Laodelphax striatella</i>	13	1	4	6
<i>Cicadula sp. female</i>	0	0	0	6
<i>Javesella dubia</i>	0	0	0	3
<i>Allygus mixtus female</i>	1	0	0	1
<i>Emelyanoviana mollicula</i>	0	2	0	0
<i>Jassargus sp. female</i>	0	0	0	1
<i>Eupelix cuspidata</i>	1	0	0	0
<i>Megamelodes quadrimaculatus</i>	0	0	1	0
Total no. of species	12	12	13	14
Total no. of individuals	326	209	163	212

Aquatic and terrestrial litter decomposition

Table S11. Overview of microbial ($k_{\text{microbial}}$) and shredder-mediated (k_{shredder}) leaf litter decomposition rates determined in the twelve FPMs. Please note that the sampling date indicates the end of the three-week sampling period.

$K_{\text{microbial}}$	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
30 May 18	0.0257	0.0284	0.0187	0.0177	0.0223	0.0242	0.0204	0.0190	0.0236	0.0246	0.0281	0.0282
28 June 18	0.0497	0.0473	0.0357	0.0496	0.0383	0.0286	0.0344	0.0286	0.0462	0.0373	0.0309	0.0293
31 July 18	0.0679	0.0585	0.0530	0.0657	0.0521	0.0627	0.0603	0.0511	0.0643	0.0397	0.0492	0.0647
24 August 18	0.0813	0.0383	0.0469	0.0447	0.0536	0.0508	0.0365	0.0444	0.0441	0.0390	0.0488	0.0368
21 September 18	0.0323	0.0350	0.0221	0.0307	0.0361	0.0391	0.0365	0.0357	0.0329	0.0300	0.0294	0.0303
19 November 18	0.0124	0.0117	0.0122	0.0089	0.0114	0.0093		0.0086	0.0051	0.0138	0.0159	0.0087
10 January 19	0.0119	0.0101	0.0153	0.0096	0.0123	0.0158	0.0160	0.0158	0.0152	0.0065	0.0081	0.0144
03 May 19	0.0173	0.0159	0.0165	0.0229	0.0200	0.0210	0.0184	0.0196	0.0179	0.0211	0.0180	0.0193
28 June 19	0.0178	0.0138	0.0235	0.0203	0.0201	0.0242	0.0239	0.0219	0.0193	0.0216	0.0246	0.0240
Overall mean	0.0351	0.0288	0.0271	0.0300	0.0296	0.0306	0.0308	0.0272	0.0298	0.0260	0.0281	0.0284
k_{shredder}	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
30 May 18	0.0140	0.0156	0.0200	0.0206	0.0139	0.0096	0.0141	0.0129	0.0121	0.0100	0.0086	0.0097
28 June 18	0.0121	0.0126	0.0156	0.0114	0.0143	0.0211	0.0158		0.0121	0.0156		0.0175
31 July 18	0.0106	0.0114	0.0136	0.0096	0.0127	0.0100	0.0107	0.0123	0.0105	0.0175	0.0116	0.0112
24 August 18	0.0043	0.0118	0.0105	0.0101	0.0084	0.0103	0.0126	0.0093	0.0116	0.0119	0.0087	0.0140
21 September 18	0.0135	0.0120	0.0149	0.0124	0.0084	0.0098	0.0110	0.0113	0.0108	0.0136	0.0129	0.0118
19 November 18	0.0045	0.0094	0.0105	0.0120	0.0114	0.0040		0.0103	0.0152	0.0100	0.0091	0.0123
10 January 19	0.0109	0.0116	0.0119	0.0121	0.0112	0.0113	0.0108	0.0106	0.0111	0.0131	0.0123	0.0112
03 May 19	0.0104	0.0119	0.0115	0.0118	0.0111	0.0111	0.0110	0.0100	0.0105	0.0102	0.0095	0.0134
28 June 19	0.0136	0.0159	0.0140	0.0143	0.0102	0.0114	0.0129	0.0114	0.0118	0.0135	0.0106	0.0110
Overall mean	0.0104	0.0125	0.0136	0.0127	0.0113	0.0109	0.0124	0.0110	0.0118	0.0128	0.0104	0.0125

Table S12. Overview of the terrestrial litter decomposition expressed as linear weight loss per day (mg/day) for green tea and rooibos tea at the FPM banks. Please note that the sampling date indicates the end of the three-week sampling period.

Green tea	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
12 December 17	33.86	23.80	27.99	29.84	35.61	28.96	28.48	30.54	28.78	30.37	33.17	27.73
21 February 18	20.37	33.63	29.71	27.78	29.51	27.45	25.57	27.85	22.16	24.02	20.85	29.23
04 April 18	29.30	29.62	31.48	32.75	33.68	29.07	29.95	32.53	32.25	28.17	32.64	27.73
28 June 18	44.01	45.08	43.30	41.90	42.67	43.83	45.16	45.09	44.31	47.82	43.73	41.57
02 August 18	19.91	27.27	28.31	17.97	27.94	27.76	25.57	27.36	30.41	34.03	32.78	32.12
27 August 18	40.68	42.79	43.47	48.62	51.70	45.56	43.62	47.79	41.79	43.36	43.74	42.81
19 September 18	48.97		50.88	47.35	51.67	49.76	43.60	50.68	47.39	48.56	41.22	50.96
19 December 18	27.65	31.34	29.50	29.17	28.86	28.85	28.09	31.71	30.61	27.82	28.97	34.06
14 January 19	28.33	28.99	27.34	29.44	30.14	29.63	27.99	38.09	31.72	31.75	27.76	29.01
12 March 19	25.64	24.96	23.84	24.50	27.85	25.21	24.70	26.84	27.50	24.60	25.55	25.62
03 May 19	34.54	36.30	37.52	35.25	37.21	36.49	36.08	37.31	37.39	34.32	32.98	34.35
22 June 19	41.24	41.90	40.78	43.44	42.53	34.98	41.61	41.58	41.47	37.50	39.35	39.69
Overall mean	32.87	33.24	34.51	34.00	36.61	33.96	33.37	36.45	34.65	34.36	33.56	34.57
Rooibos tea	FPM1	FPM2	FPM3	FPM4	FPM5	FPM6	FPM7	FPM8	FPM9	FPM10	FPM11	FPM12
12 December 17	17.11	17.60	16.21	20.34	17.58	16.82	20.18	18.18	16.82	16.38	15.02	16.57
21 February 18	15.60	20.55	18.58	18.37	17.67	17.99	19.84		14.77	17.57	14.45	14.02
04 April 18	20.42	20.26	20.11	20.44	19.96	21.20	19.06	18.30	21.88	19.18	20.43	21.50
28 June 18	21.37	25.66	26.05	20.12	19.77	24.89	20.01	29.20	25.99	25.67	27.59	26.60
02 August 18	9.84	13.41	9.90	9.95	12.22	19.58	8.92	14.44	15.06	18.88	18.62	16.39
27 August 18	16.70	20.33	23.47	18.09	19.15	19.47	19.29	20.57	21.41	27.66	28.31	28.04
19 September 18	16.85	19.55	12.30	15.55	16.68	23.37	22.49	19.71	22.49	23.07	18.62	20.18
19 December 18	9.01	9.84	10.71	11.11	12.33	15.27	11.15	16.04	12.54	12.27	10.67	9.89
14 January 19	12.95	12.06	10.98	11.40	12.22	10.99	11.98	12.77	13.15	13.37	10.54	12.63
12 March 19	13.02	13.17	12.65	14.12	12.74	14.55	12.24	13.76	11.56	15.62	13.83	13.94
03 May 19	19.21	19.57	16.45	18.97	16.68	19.29	14.60	20.28	16.75	20.18	21.69	18.73
22 June 19	24.46	18.37	21.97	31.91	23.72	21.32	25.93	24.66	28.58	22.16	24.62	25.15
Overall mean	16.38	17.53	16.61	17.53	16.73	18.73	17.14	18.90	18.42	19.33	18.70	18.64

Principal component analyses (PCA) and redundancy analyses (RDA)

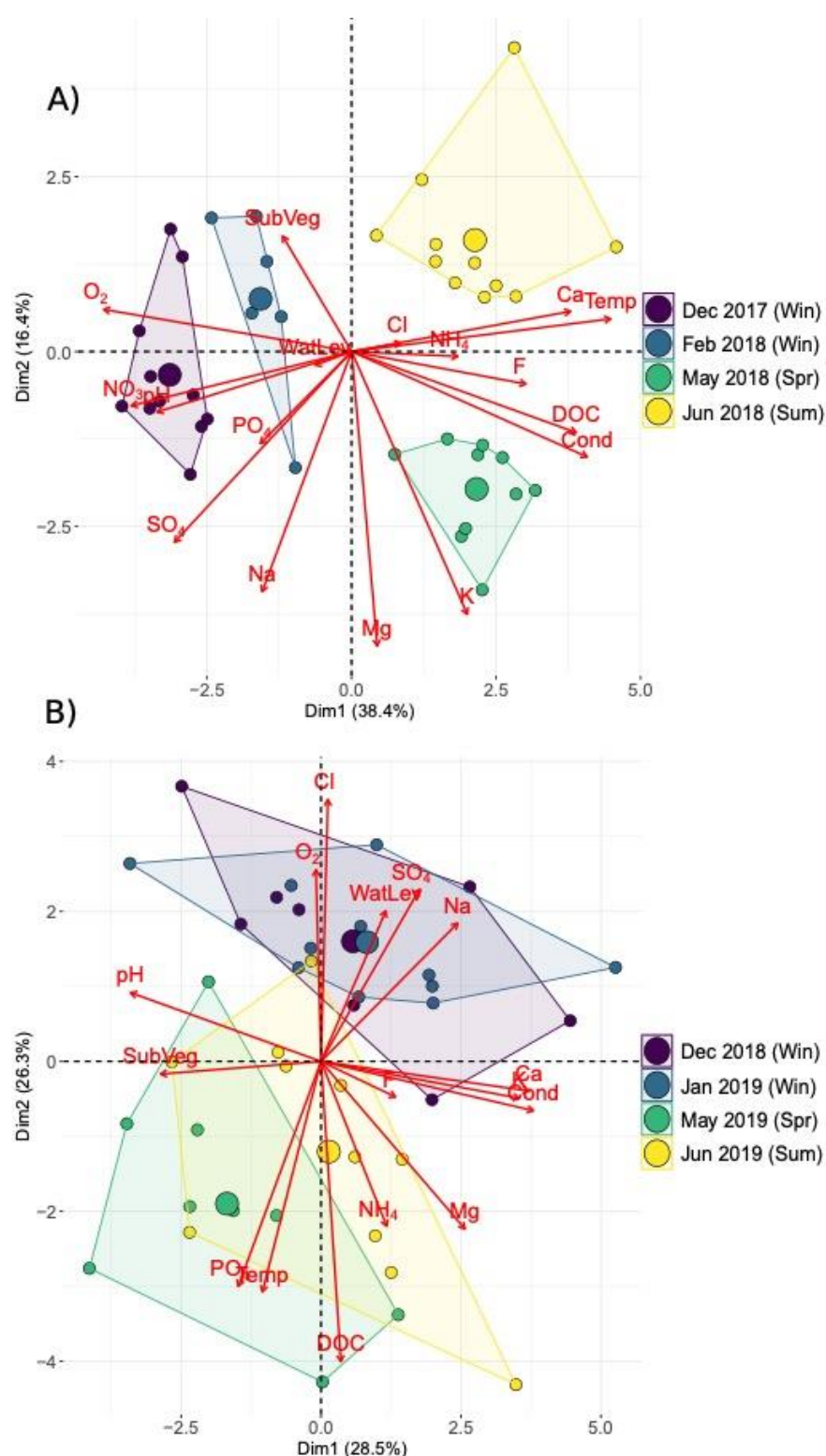


Figure S9. PCA biplots of ecosystem development and environmental conditions in the twelve FPMs for (A) winter 2017/2018 till summer 2018 and (B) winter 2018/2019 till summer 2019. Small dots represent FPMs and coloured polygons represent respective months and seasons (see legend for details), with larger dots representing the mean for a given month.

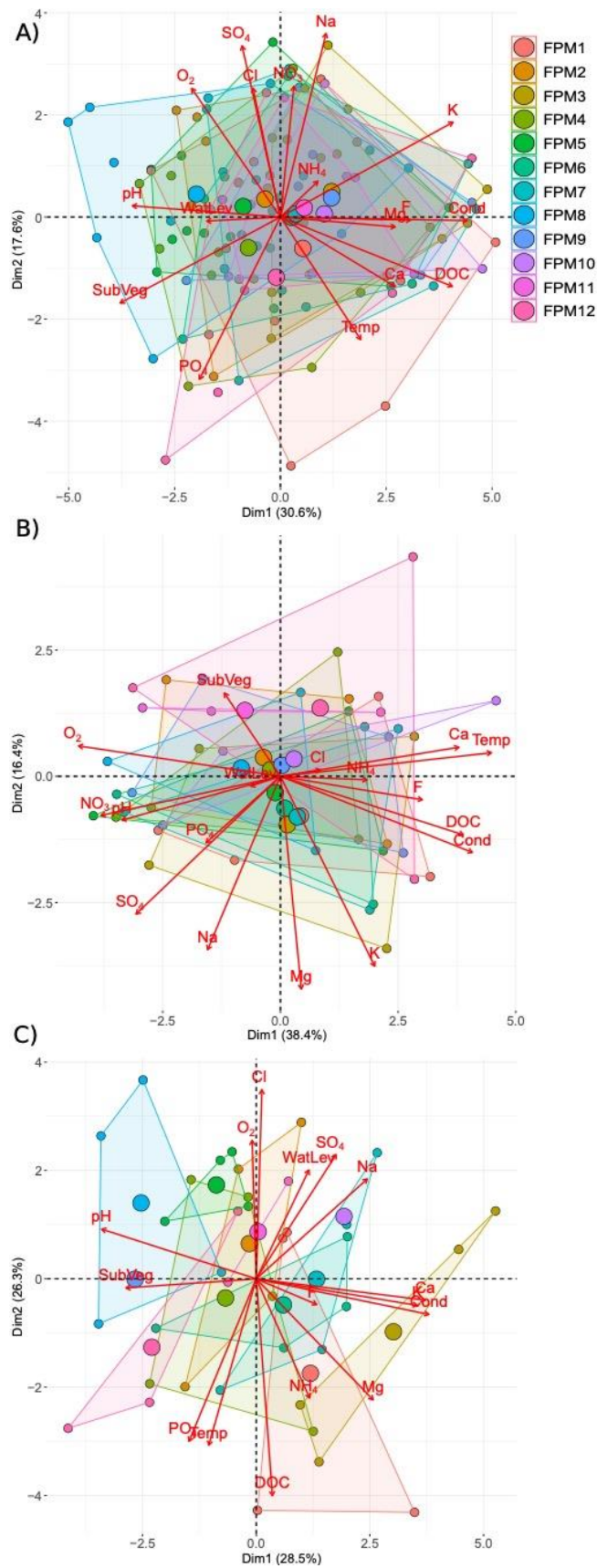


Figure S10. PCA biplots of environmental conditions during the individual seasons grouped by the twelve FPMs, with (A) showing available data for the entire study period (i.e., December 2018 – June 2019), (B) showing data for the first year (i.e., December 2018 – June 2019) and (C) showing data for the second year (i.e., December

2018 till June 2019) of ecosystem developments. Small dots represent individual months and coloured polygons represent respective FPMs (see legend for details), with larger dots representing the mean for a given FPM.

Table S13. Principal component (PC1, PC2) loadings for the principal component analysis.

PC1	Loading	PC2	Loading
Specific conductivity	0.91	Sodium	0.74
Potassium	0.84	Sulfate	0.69
DOC	0.84	Nitrate	0.53
Fluoride	0.64	Chloride	0.52
Magnesium	0.56	Dissolved oxygen	0.52
Calcium	0.56	Potassium	0.38
Water temperature	0.39	DOC	- 0.28
Sodium	0.22	Calcium	- 0.28
Sulfate	- 0.19	Submerged vegetation	- 0.35
Water level	- 0.36	Water temperature	- 0.50
Phosphate	- 0.40	Phosphate	- 0.66
Dissolved oxygen	- 0.43		
pH	- 0.72		
Submerged vegetation	- 0.78		

The following variables were excluded from principal component analyses (PCA) and redundancy analyses (RDA): Soil temperature, TOC, benthic macroinvertebrates, merolimnic insect emergence, zooplankton, tadpole coverage, crayfish and fish, terrestrial vegetation, ground beetles, leafhoppers, aquatic and terrestrial litter decomposition (see also methods section).

Table S14. Results of the redundancy analysis (RDA) for the entire study period (R^2 adj. = 0.777).

	Df	Variance	F	p adj. (Bonferroni)
Pond	11	0.049	2.049	0.014
Season	2	0.013	3.103	0.017
Time (month) from beginning	1	0.02	9.198	0.003
Residuals	62	0.135		

Table S15. Results of the redundancy analysis (RDA) for the first year of ecosystem development (R^2 adj. = 0.556).

	Df	Variance	F	p adj. (Bonferroni)
Season	2	0.054	6.041	0.002
Time (month) from beginning	1	0.02	4.412	0.006
Residuals	36	0.162		

Table S16. Results of the redundancy analysis (RDA) for the second year of ecosystem development (R^2 adj. = 0.698).

	Df	Variance	F	p adj. (Bonferroni)
Pond	11	0.0869	3.639	0.003
Season	2	0.04345	10.003	0.003
Time (month) from beginning	1	0.00633	2.9153	0.023
Residuals	24	0.05212		